

Becker's The World of the Cell, 9e (Hardin/Bertoni/Kleinsmith)
Chapter 13 Cytoskeletal Systems

13.1 Multiple-Choice Questions

1) The _____ is a complex network of interconnected filaments and tubules that extends throughout the cytosol from the nucleus to the inner surface of the plasma membrane.

- A) cytoskeleton
- B) microtubule apparatus
- C) axoneme
- D) desmin sheath
- E) microfilament array

Answer: A

Chapter Section: 13.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

2) Cytoskeletal elements are found in

- A) bacteria.
- B) archaea.
- C) eukaryotes.
- D) B and C only.
- E) A, B, and C.

Answer: E

Chapter Section: 13.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

3) Which of the following is *not* a function of the cytoskeleton?

- A) cell locomotion
- B) cell division
- C) organelle movement
- D) establishing cell shape
- E) passive transport

Answer: E

Chapter Section: 13.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

4) Which of the following cytoskeletal proteins is found in prokaryotes?

- A) keratin
- B) tubulin
- C) crescentin
- D) vimentin
- E) actin

Answer: C

Chapter Section: 13.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

5) Microfilaments function in

- A) muscle contraction.
- B) cytoplasmic streaming.
- C) cell division.
- D) maintenance of animal cell shape.
- E) All of the above.

Answer: E

Chapter Section: 13.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

6) Microtubules have an outer diameter of

- A) 7 nm.
- B) 8—12 nm.
- C) 25 nm.
- D) 7 mm.
- E) 25 mm.

Answer: C

Chapter Section: 13.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

7) Microtubules contribute to each of the following *except*

- A) intracellular transport.
- B) the human red blood cell disklike shape.
- C) orientation of plant cellulose microfibrils.
- D) filopodial structure.
- E) mitotic spindles.

Answer: D

Chapter Section: 13.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

8) Intermediate filaments

- A) are 7 nm in diameter.
- B) can be composed of the protein vimentin.
- C) are abundant in plant cells.
- D) are the only cytoskeletal filament responsible for animal cell shape.
- E) A, B, C, and D.

Answer: B

Chapter Section: 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

9) Which technique would best be used to visualize movement of unlabeled vesicles along microtubules?

- A) immunofluorescence microscopy of fixed material
- B) live cell fluorescence microscopy
- C) computer-enhanced digital video microscopy
- D) deep-etch electron microscopy
- E) A, B, C, or D would be equally useful.

Answer: C

Chapter Section: 13.1

Bloom's Taxonomy: Application

Learning Outcome: 13.1

Global LO: G1

10) Which of the following is *true* about the structure of the microtubule wall?

- A) α - β -tubulin heterodimers are arranged with mixed orientation in each protofilament, and protofilaments are arranged with mixed polarity in the microtubule wall.
- B) α - β -tubulin heterodimers are arranged with the same orientation in each protofilament, and protofilaments are arranged with mixed polarity in the microtubule wall.
- C) α - β -tubulin heterodimers are arranged with the mixed orientation in each protofilament, and protofilaments are arranged with same polarity in the microtubule wall.
- D) α - β -tubulin heterodimers are arranged with the same orientation in each protofilament, and protofilaments are arranged with same polarity in the microtubule wall.
- E) Orientation of the α - β -tubulin heterodimers is completely random and depends on which end of the dimer first hits the end of the microtubule when it assembles.

Answer: D

Chapter Section: 13.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

11) Which phase of microtubule polymerization *in vitro* would be affected most by adding short microtubules to unpolymerized GTP-tubulin?

- A) lag phase
- B) elongation phase
- C) addition of tubulin dimers to microtubule (+) ends at the plateau phase
- D) addition of tubulin dimers to microtubule (−) ends at the plateau phase
- E) treadmilling

Answer: A

Chapter Section: 13.2

Bloom's Taxonomy: Evaluation

Learning Outcome: 13.1

Global LO: G2

12) Which of the following is *false* about dynamic instability of a microtubule end?

- A) During a growth phase, assembly of GTP-tubulin is faster than incorporated GTP is hydrolyzed, forming a GTP-cap.
- B) During a catastrophe phase, hydrolysis of incorporated GTP is faster than GTP-tubulin addition, forming a GDP-cap.
- C) During rescue, GTP-tubulin assembles fast enough to reestablish a GTP cap.
- D) Each end of a microtubule can undergo dynamic instability independently.
- E) A, B, C, and D are true.

Answer: E

Chapter Section: 13.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

13) Tubulin may assemble onto one end while disassembling from the other end of the same microtubule simultaneously in a process known as

- A) dynamic instability.
- B) treadmilling.
- C) nucleation.
- D) oligomerization.
- E) catastrophe-rescue.

Answer: B

Chapter Section: 13.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

- 14) The critical concentration for tubulin polymerization is the concentration of
- A) tubulin dimers at which all dimers assemble into microtubules.
 - B) unassembled tubulin dimers when assembly is balanced with disassembly.
 - C) tubulin dimers at which dimers assemble onto microtubule minus ends but disassemble from plus ends.
 - D) unassembled tubulin dimers when there is net disassembly of dimers from both microtubule ends.
 - E) tubulin dimers in most cells.

Answer: B

Chapter Section: 13.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

- 15) Dynamic assembly and disassembly of a cytoplasmic microtubule in a cell occurs primarily at its plus end, because its minus end is usually anchored to

- A) a basal body.
- B) a centromere.
- C) a microtubule organizing center.
- D) a mitochondrial outer membrane.
- E) the plasma membrane.

Answer: C

Chapter Section: 13.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

- 16) Axonemal microtubules are

- A) dynamically unstable.
- B) anchored on basal bodies.
- C) randomly oriented in the axoneme.
- D) required to maintain axons.
- E) involved in intracellular organelle transport.

Answer: B

Chapter Section: 13.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

17) _____ causes microtubules to form tight bundles in nerve cell axons.

- A) MAP2
- B) EB1
- C) Tau
- D) FtsZ
- E) Desmin

Answer: C

Chapter Section: 13.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

18) Which of the following proteins is associated with the precise control of where microtubules depolymerize during mitosis?

- A) Tau
- B) stathmin/Op18
- C) gelsolin
- D) catastrophins
- E) MAP 2

Answer: D

Chapter Section: 13.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

19) CapZ

- A) causes microfilaments to lengthen by assembly at their plus ends.
- B) depolymerizes microfilaments at their plus ends.
- C) prevents assembly and disassembly at microfilament plus ends.
- D) forms microfilament branches.
- E) connects microfilaments to microtubules.

Answer: C

Chapter Section: 13.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

20) Intermediate filaments

- A) lack the intrinsic polarity of microtubules and microfilaments, because IF tetrameric subunits are symmetric.
- B) are grouped into five cytoplasmic classes based on the cell types in which they are found and one nuclear class.
- C) are the most stable and least soluble cytoskeleton filaments.
- D) are not found in plant cell cytoplasm.
- E) All of the above are true.

Answer: E

Chapter Section: 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G7

21) Centrosome ring-shaped complexes contain _____ and _____.

- A) α -tubulin; β -tubulin
- B) γ -tubulin; MCAK
- C) γ -tubulin; MAP2
- D) γ -tubulin; GRiPs
- E) γ -tubulin; crescentin

Answer: D

Chapter Section: 13.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

22) What is the effect of cofilin on an actin microfilament?

- A) caps the microfilament plus end
- B) caps the microfilament minus end
- C) increases turnover of ATP-actin at the microfilament plus end
- D) increases turnover of ADP-actin at the microfilament plus end
- E) increases turnover of ADP-actin at the microfilament minus end

Answer: E

Chapter Section: 13.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

23) A microtubule-organizing center (MTOC)

- A) contains centrioles in plant cells.
- B) serves as a site for both microtubule and intermediate filament assembly.
- C) anchors either end of cytoplasmic microtubules.
- D) anchors only the minus ends of cytoplasmic microtubules.
- E) usually is positioned near the cell membrane.

Answer: D

Chapter Section: 13.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

24) _____ is a drug that blocks cell division by stabilizing microtubules; as a result, it is used in the treatment of _____.

- A) Vincristine; listeriosis
- B) Vinblastine; lung cancer
- C) Taxol; breast cancer
- D) Nocodazole; athlete's foot
- E) Colchicine; gout

Answer: C

Chapter Section: 13.2

Bloom's Taxonomy: Application

Learning Outcome: 13.1

Global LO: G5

25) Which of the following proteins would be a good drug target for interfering with microfilament network breakdown?

- A) profilin
- B) CapZ
- C) gelsolin
- D) tropomodulin
- E) calmodulin

Answer: C

Chapter Section: 13.3

Bloom's Taxonomy: Evaluation

Learning Outcome: 13.1

Global LO: G2

26) Microfilaments

- A) are composed of eight protofilaments joined end to end with staggered overlaps.
- B) are composed of two helical strands of actin monomers oriented in the same direction.
- C) are composed of two helical antiparallel strands of actin monomers.
- D) are composed of α - and β -actin dimers.
- E) polymerize at the same rate from both ends of the filament.

Answer: B

Chapter Section: 13.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

27) Actin filaments may form each of the following *except*

- A) 3D filament networks (gels).
- B) contractile bundles.
- C) branched networks.
- D) bundles of parallel filaments.
- E) axonemes containing doublet microfilaments.

Answer: E

Chapter Section: 13.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

28) Which of the following intermediate filament proteins is incorrectly matched with the cell type in which it is found?

- A) keratin - epithelial cells
- B) desmin - smooth muscle cells
- C) vimentin - mesenchymal cells
- D) nestin - connective tissue cells
- E) GFA protein - glial cells

Answer: D

Chapter Section: 13.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 13.1

Global LO: G2

29) Which of the following is *not* known to be involved in associating actin microfilaments with plasma membranes?

- A) PIP2
- B) spectrin
- C) ankyrin
- D) band 4.1
- E) A, B, C, and D are involved in associating actin microfilaments with plasma membranes.

Answer: A

Chapter Section: 13.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

30) Profilin does all of the following *except*

- A) competes with thymosin β 4 for binding to actin monomers.
- B) sequesters actin monomers to prevent any polymerization.
- C) binds to formin in order to "stage" actin monomers for addition to a growing filament.
- D) promotes actin monomer addition to a filament plus end.
- E) Profilin does all of the above.

Answer: B

Chapter Section: 13.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

31) Which of the following is *not* associated with the bundle of microfilaments supporting each microvillus in small intestine mucosal cells?

- A) fimbrin
- B) villin
- C) fascin
- D) myosin I
- E) All are associated with the microvillus microfilament bundle.

Answer: C

Chapter Section: 13.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

32) Mutations of keratins in the _____ that resist tensile forces in the skin cause the blistering disease known as epidermolysis bullosa simplex.

- A) intermediate filaments
- B) microtubules
- C) microfilaments
- D) actin filaments
- E) crescentin filaments

Answer: A

Chapter Section: 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G7

33) Actin bundles give shape and support to _____ found in intestinal mucosal cells.

- A) cilia
- B) flagella
- C) microvilli
- D) nuclei
- E) lamellipodia

Answer: C

Chapter Section: 13.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

34) Activation of which of the following is associated with the formation of focal adhesions and stress fibers?

- A) Cdc42
- B) plectin
- C) Rho
- D) Rac
- E) γ -tubulin

Answer: C

Chapter Section: 13.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

35) Rho Family GTPases (Rho, Rac, Cdc42) stimulate cytoskeletal microfilament assembly for all of the following *except*

- A) stress fiber formation.
- B) lamellipodia formation.
- C) nuclear envelope breakdown for mitosis.
- D) cytokinetic furrow formation.
- E) filopodia formation.

Answer: C

Chapter Section: 13.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

36) Which of the following is *false* about lamellipodia?

- A) Their formation is stimulated by activation of Rac.
- B) They may have associated filopodia.
- C) They contain bundles of microtubules that push the plasma membrane forward.
- D) They extend at the leading edge of crawling cells.
- E) They contain branched networks of actin filaments.

Answer: C

Chapter Section: 13.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

37) Microtubules and microfilaments are *similar* with respect to each of the following *except*

- A) inherent polarity of the structure.
- B) localization similar to that of intermediate filaments in cells.
- C) association with motor proteins.
- D) plus and minus ends for polymerization.
- E) both are polymers of subunits.

Answer: B

Chapter Section: 13.2

Bloom's Taxonomy: Analysis

Learning Outcome: 13.1

Global LO: G7

38) All of the following are components associated with bacterial cytoskeletons *except*

- A) FtsZ.
- B) MreB.
- C) plectin.
- D) crescentin.
- E) A, B, C, and D are associated with the bacterial cytoskeletons.

Answer: C

Chapter Section: 13.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

39) Which of the following cytoskeletal proteins would be expected to be involved in binary fission of a bacterium into two bacteria?

- A) FtsZ
- B) MreB
- C) plectin
- D) crescentin
- E) tubulin

Answer: A

Chapter Section: 13.1

Bloom's Taxonomy: Evaluation

Learning Outcome: 13.1

Global LO: G2

40) Which of the following microtubule-interacting proteins would you target to disrupt cell division in actively replicating tumor cells?

- A) Tau
- B) EB1 (a +-TIP protein)
- C) MCAK (a catastrophin)
- D) plectin (a spectraplaklin)
- E) MAP2

Answer: C

Chapter Section: 13.2

Bloom's Taxonomy: Evaluation

Learning Outcome: 13.1

Global LO: G2

13.2 Matching Questions

Choose the item in column 2 that best matches each item in column 1.

- A) increase cell surface area
- B) crosslinks microtubules to intermediate filaments
- C) nucleates actin filament branches
- D) decorates actin microfilaments
- E) connects crisscrossing actin filaments into 3D networks
- F) forms scaffold underlying nuclear envelope
- G) formation of tubulin oligomers
- H) stabilizes and organizes microtubules
- I) breaks and caps actin filaments
- J) assembly and disassembly on opposite ends of the same filament
- K) is in cilia and flagella
- L) nucleates microtubule branches
- M) curves bacterial cells
- N) prevents loss of subunits from filament minus end
- O) keeps muscle myofibrils in register
- P) inhibits microtubule assembly
- Q) anchors spectrin filaments to red blood cell membrane
- R) binds to profilin and CapZ
- S) desmosome
- T) hemidesmosome
- U) binds to microtubule plus ends

1) axonemal microtubules

Chapter Section: 13.2, 13.3, 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

2) nucleation

Chapter Section: 13.2, 13.3, 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

3) treadmilling

Chapter Section: 13.2, 13.3, 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

4) desmin

Chapter Section: 13.2, 13.3, 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

5) EB1

Chapter Section: 13.2, 13.3, 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

6) Arp2/3 complex

Chapter Section: 13.2, 13.3, 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

7) myosin subfragment 1 (S1)

Chapter Section: 13.2, 13.3, 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

8) tropomodulin

Chapter Section: 13.2, 13.3, 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

9) gelsolin

Chapter Section: 13.2, 13.3, 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

10) microvilli

Chapter Section: 13.2, 13.3, 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

11) filamin

Chapter Section: 13.2, 13.3, 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

12) phosphoinositides

Chapter Section: 13.2, 13.3, 13.4
Bloom's Taxonomy: Knowledge
Learning Outcome: 13.1
Global LO: G2

13) MAPs
Chapter Section: 13.2, 13.3, 13.4
Bloom's Taxonomy: Knowledge
Learning Outcome: 13.1
Global LO: G2

14) plectin
Chapter Section: 13.2, 13.3, 13.4
Bloom's Taxonomy: Knowledge
Learning Outcome: 13.1
Global LO: G2

15) lamin
Chapter Section: 13.2, 13.3, 13.4
Bloom's Taxonomy: Knowledge
Learning Outcome: 13.1
Global LO: G2

Answers: 1) K 2) G 3) J 4) O 5) U 6) C 7) D 8) N 9) I 10) A 11) E 12) R 13) H 14) B 15) F

13.3 Short Answer Questions

1) The _____ is a dense network of microfilaments just beneath the plasma membrane.

Answer: cell cortex

Chapter Section: 13.3
Bloom's Taxonomy: Knowledge
Learning Outcome: 13.1
Global LO: G2

2) Based on composition similarities, actin can be divided into two major groups: the muscle-specific actins (_____ -actin) and nonmuscle actins (_____ - and _____ -actin).

Answer: alpha; beta; gamma

Chapter Section: 13.3
Bloom's Taxonomy: Knowledge
Learning Outcome: 13.1
Global LO: G2

3) The Mitchison and Kirschner _____ model of how microtubules transition between periods of growth and catastrophe proposes that _____ bound to β -tubulin subunits at a microtubule plus end stabilizes the end and promotes growth, but if the rate of hydrolysis to _____ exceeds the rate of tubulin addition, the end becomes unstable and loses subunits.

Answer: dynamic instability; GTP; GDP

Chapter Section: 13.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

4) γ -TuRCs contain _____-tubulin and nucleate microtubule growth from MTOCs known as _____.

Answer: γ ; centrosomes

Chapter Section: 13.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

5) Activation of the _____ G-protein results in the extension of lamellipodia.

Answer: Rac

Chapter Section: 13.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

6) The core of an intestinal microvillus is composed of a tight bundle of _____.

Answer: microfilaments

Chapter Section: 13.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

7) A network of _____, a "fourth cytoskeleton" is involved in pinching off yeast daughter cells.

Answer: septins

Chapter Section: 13.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

8) A unique feature of _____ is that they are tissue-specific.

Answer: intermediate filaments

Chapter Section: 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

9) Cytochalasins prevent addition of monomers to preexisting _____.

Answer: microfilaments

Chapter Section: 13.3
Bloom's Taxonomy: Knowledge
Learning Outcome: 13.1
Global LO: G2

10) The two main types of eukaryotic microtubules are the _____ and _____ microtubules.

Answer: cytoplasmic; axonemal

Chapter Section: 13.2
Bloom's Taxonomy: Knowledge
Learning Outcome: 13.1
Global LO: G2

11) _____ proteins promote depolymerization of existing microtubules at the ends.

Answer: Catastrophin

Chapter Section: 13.2
Bloom's Taxonomy: Knowledge
Learning Outcome: 13.1
Global LO: G2

13.4 Inquiry

Many drugs have the ability to disrupt the cytoskeletal structure of the cell. These drugs are useful in the study of microtubules and microfilaments.

1) What are some of the drugs used to study microfilaments?

Answer: cytochalasin D, latrunculin A, phalloidin

Chapter Section: 13.3
Bloom's Taxonomy: Knowledge
Learning Outcome: 13.1
Global LO: G2

2) What are some of the drugs used to study microtubules?

Answer: colchicine, vincristine, vinblastine, nocodazole, taxol

Chapter Section: 13.2
Bloom's Taxonomy: Knowledge
Learning Outcome: 13.1
Global LO: G2

3) Describe an additional experimental approach used to study the cytoskeleton in which the activity of cytoskeletal elements is affected in ways similar to those resulting from drug treatment.

Answer: Using organisms or cell lines expressing mutant cytoskeletal elements reveals effects of their inhibition or over activation on cell activities.

Chapter Section: 13.2, 13.3, 13.4

Bloom's Taxonomy: Application

Learning Outcome: 13.1

Global LO: G2

4) Cancer is a disease in which cells exhibit uncontrolled cell division. Drugs used in the study of microtubules are also used in the treatment of cancer. Why are drugs that disrupt microtubules more useful in the treatment of cancer than are drugs used to disrupt microfilaments? What is the name of the category of these drugs that describes their function?

Answer: Microtubules, especially those in the mitotic spindle, are more important than microfilaments in mitosis. Uncontrolled division of cancer cells depends on the correct functioning of microtubules. Drugs that disrupt the mitotic spindle are categorized as antimitotic drugs.

Chapter Section: 13.2

Bloom's Taxonomy: Evaluation

Learning Outcome: 13.1

Global LO: G8

The proteins of intermediate filaments (IFs) are tissue-specific. Unlike microtubules or microfilaments, the amino acid composition of intermediate filaments varies from tissue to tissue. IFs are classified by the cell type or tissue in which they are found.

5) According to your text, how many classes of IFs exist?

Answer: six

Chapter Section: 13.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 13.1

Global LO: G2

6) In the table below, list the classes of IF, the IF protein subunits, and the tissue (or cells) in which they are found.

Class	IF Protein	Tissue/Cells

Answer: See both Table 13-4 and the table below.

Class	IF Protein	Tissue/Cells
I	Acidic cytokeratins	Epithelial
II	Basic cytokeratins	Epithelial
III	Vimentin, desmin, GFA protein	Fibroblasts; cells of mesenchymal origin; eye lens; muscle cells; glial; astrocytes
IV	Neurofilament proteins	Nerves
V	Nuclear lamins	All cell types
VI	Nestin	Neuronal stem cells

Chapter Section: 13.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 13.1

Global LO: G2

7) Cancer cells retain the IF proteins from the tissue of origin even after metastasizing to other tissues. How is this important in the diagnosis and treatment of cancer?

Answer: Evidence for metastasis (movement of cancer cells to other tissues) of a tumor usually indicates a poorer prognosis. Treatment strategies are based on the cancer cell origin, which is revealed by determining the type of intermediate filaments expressed.

Chapter Section: 13.4

Bloom's Taxonomy: Application

Learning Outcome: 13.1

Global LO: G8

8) Explain how the discovery of cytoskeletal elements in prokaryotes contributed to the current understanding of the endosymbiont theory.

Answer: The tubulin-like protein FtsZ is involved in bacterial cell division. Expression of FtsZ in chloroplasts and some mitochondria and its involvement in chloroplast division is evidence that chloroplasts and mitochondria originally were endosymbionts in the original eukaryotic cells.

Chapter Section: 13.1

Bloom's Taxonomy: Synthesis

Learning Outcome: 13.1

Global LO: G8

9) You have discovered a new cytoskeletal element and need to identify and characterize it for

further work. How would you go about examining this protein to ultimately classify it as a cytoskeletal element?

Answer: First, isolate the subunit and determine whether it polymerizes and has a protein sequence similar to other known cytoskeletal proteins. Second, determine whether the subunit has structural polarity and structural similarity to other cytoskeletal proteins. Third, determine whether function is affected by a variety of known cytoskeletal protein inhibitors.

Chapter Section: 13.1

Bloom's Taxonomy: Synthesis

Learning Outcome: 13.1

Global LO: G1

10) What is the importance of proteins that precisely control when and where cytoskeletal elements polymerize and depolymerize? Make sure to give examples in your explanations.

Answer: Their precise control of filament existence, location, orientation, length, and stability is required for the wide variety of cell structures and activities supported by the three types of cytoskeletal filaments. Examples include catastrophe activity associated with spindle fiber microtubules, microfilament formation of actin filaments by formin in stress fibers and by Arp2/3 in lamellipodia, etc.

Chapter Section: 13.2, 13.3, 13.4

Bloom's Taxonomy: Application

Learning Outcome: 13.1

Global LO: G7

Becker's The World of the Cell, 9e (Hardin/Bertoni/Kleinsmith)
Chapter 14 Cellular Movement: Motility and Contractility

14.1 Multiple-Choice Questions

1) Which of the following statements is correct with respect to flagella and cilia?

- A) The flagellum is usually at the leading end of a swimming sperm cell.
- B) Cilia generate a force parallel to the cell surface.
- C) There usually are fewer cilia than flagella.
- D) Cilia generally are longer than flagella.
- E) Cilia often have a helical pattern of movement.

Answer: B

Chapter Section: 14.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1

Global LO: G2

2) The axonemal "9 + 2" pattern of microtubules is nine _____ surrounding a(n) _____.

- A) outer doublets; central pair
- B) basal bodies; inner two
- C) central tubules; double pair
- D) B tubule structures; middle two
- E) triplet microtubules; basal bodies

Answer: A

Chapter Section: 14.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1

Global LO: G2

3) The "9 + 0" pattern of microtubules is associated with

- A) bacterial locomotion.
- B) cyanobacterial locomotion.
- C) respiratory tract cilia.
- D) sperm tail flagella.
- E) primary (sensory) cilia.

Answer: E

Chapter Section: 14.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1

Global LO: G2

4) Which of the following lists of structures is in the correct order from inside to outside of the cell?

- A) axoneme, transition zone, basal body
- B) transition zone, basal body, axoneme
- C) axoneme, basal body, transition zone
- D) basal body, transition zone, axoneme
- E) basal body, axoneme, transition zone

Answer: D

Chapter Section: 14.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1

Global LO: G7

5) In the axoneme, dynein sidearms are anchored on

- A) B tubules.
- B) A tubules.
- C) central pair tubules.
- D) radial spokes.
- E) interdouplet nexin.

Answer: B

Chapter Section: 14.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1

Global LO: G7

6) Each of the following diseases is classified as a primary cilia dysplasia (also known as a ciliopathy) *except*

- A) Kartagener's syndrome.
- B) situs inversus totalis.
- C) heterotaxia.
- D) Griscelli's disease.
- E) All are primary cilia dysplasias.

Answer: D

Chapter Section: Human Connections

Bloom's Taxonomy: Comprehension

Learning Outcome: 14.1

Global LO: G5

7) _____ links the thick filaments to the Z line.

- A) Troponin
- B) Tropomyosin
- C) Titin
- D) Nebulin
- E) Myomesin

Answer: C

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

8) Striated muscle has light and dark bands. The light bands are called _____, whereas the dark bands are called _____.

- A) Z lines; M lines
- B) H zones; M zones
- C) A bands; I bands
- D) I bands; A bands
- E) M lines; A bands

Answer: D

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

9) Cytological examination of resting and contracted muscle provides important support for the sliding filaments model for how skeletal muscle contracts. This evidence includes the observation that during contraction the

- A) Z lines move farther apart.
- B) lengths of the thick filaments shorten significantly.
- C) lengths of the thin filaments shorten significantly.
- D) I bands get longer.
- E) H zone becomes narrower.

Answer: E

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

10) Biochemical evidence supporting the sliding filament model of muscle contraction includes the observation that

- A) myosin light chains can be phosphorylated.
- B) titin connects each end of a thick filament to a Z line.
- C) TnC binds Ca^{2+} .
- D) actin has polymerization ATPase activity.
- E) myosin has an ATP-dependent crossbridge cycle.

Answer: E

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

11) Which of the following major protein components of skeletal muscle is *incorrectly* matched with its function?

- A) titin—links thick filaments to Z line
- B) nebulin—links thin filaments to Z line
- C) myomesin—links myosin bipolar filaments at M line
- D) troponin—binds along the length of the thin filaments and blocks myosin binding sites
- E) actin—major component of thin filaments

Answer: D

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

12) Motility may involve

- A) movement of a cell/organism through its environment.
- B) movement of the environment past a cell.
- C) movement of components within a cell.
- D) shortening of a cell.
- E) All of the above are correct.

Answer: E

Chapter Section: 14.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

13) Which of the following changes take place as the muscle contracts?

- A) The Z lines get closer together.
- B) The H zones get smaller.
- C) The I band gets smaller.
- D) The A band gets smaller.
- E) A, B, and C are correct.

Answer: E

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

14) In relaxed muscle, Ca^{2+} is found in highest concentration in the

- A) sarcoplasm.
- B) T tubules.
- C) sarcoplasmic reticulum.
- D) sarcolemma.
- E) neuromuscular junction.

Answer: C

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

15) All of the following are actin-dependent nonmuscle motilities *except*

- A) extension of the ER.
- B) cleavage furrow constriction in cytokinesis.
- C) chemotaxis.
- D) cyclosis or cytoplasmic streaming.
- E) amoeboid movement.

Answer: A

Chapter Section: 14.5

Bloom's Taxonomy: Analysis

Learning Outcome: 14.2

Global LO: G2

16) Which of the following characteristics is *not* associated with cardiac muscle?

- A) intercalated discs
- B) electrical coupling by gap junctions
- C) involuntary nervous control
- D) fusion of cardiomyocytes into large multinucleated myofibers
- E) striations

Answer: D

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

17) Ca^{2+} -calmodulin stimulates phosphorylation of myosin by

- A) Ca²⁺-ATPase.
- B) focal adhesion kinase.
- C) myosin light chain kinase.
- D) ryanodine receptor.
- E) titin.

Answer: C

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

18) Which of the following is *incorrect* about stimulation of striated muscle contraction by the neuromuscular junction?

- A) Acetylcholine is stored in vesicles in the axon terminal.
- B) Acetylcholine is secreted in response to a nerve action potential.
- C) Acetylcholine binds to soluble receptors in the extracellular environment.
- D) Acetylcholine binding stimulates sodium ions to enter the muscle cell.
- E) T tubules carry action potentials into the muscle cell.

Answer: C

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

19) Motor proteins convert

- A) mechanical energy to chemical energy.
- B) heat energy to mechanical energy.
- C) kinetic energy to mechanical energy.
- D) chemical energy to mechanical energy.
- E) chemical energy to heat energy.

Answer: D

Chapter Section: 14.0

Bloom's Taxonomy: Comprehension

Learning Outcome: 14.2

Global LO: G2

20) Which of the following features is *not* shared by microtubule and microfilament molecular motors?

- A) coupling ATP hydrolysis to change shape and attachment of the motor to its associated cytoskeletal filament
- B) retention of bound ADP and Pi products immediately after ATP hydrolysis
- C) cycles of ATP hydrolysis, release of ADP and Pi, and binding of new ATP as they undergo a crossbridge cycle of force production on a filament
- D) similar structural features that transduce changes in the motor protein shape into movement
- E) The motors share all these features.

Answer: E

Chapter Section: 14.0

Bloom's Taxonomy: Comprehension

Learning Outcome: 14.2

Global LO: G2

21) Arp2/3-dependent actin polymerization directly drives which of the following activities at the leading edge of crawling cells?

- A) filament rearward translocation
- B) lamellipodia protrusion
- C) new adhesion formation
- D) generation of tension
- E) focal adhesion detachment

Answer: B

Chapter Section: 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

22) Where would you expect to find the Ca²⁺-activated form of gelsolin in cells crawling by amoeboid movement?

- A) in and around focal adhesions
- B) at a pseudopodium tip where endoplasm is converting to ectoplasm
- C) at a pseudopodium base where ectoplasm is converting to endoplasm
- D) in the primary cilium
- E) at the cell trailing edge where force is produced to drive endoplasm forward streaming

Answer: C

Chapter Section: 14.5

Bloom's Taxonomy: Analysis

Learning Outcome: 14.2

Global LO: G2

23) Which of the following is the correct order of events in a crawling cell?

- A) protrusion at the cell's leading edge, attachment of protrusion, pulling the cell forward, release and retraction of trailing edge
- B) attachment of protrusion, pulling the cell forward, protrusion at the cell's leading edge, release and retraction of trailing edge
- C) release and retraction of protrusion, protrusion at the cell's leading edge, attachment of trailing edge, pulling the cell forward
- D) attachment of protrusion, protrusion at the cell's leading edge, pulling the cell forward, release and retraction of trailing edge
- E) pulling the cell forward, release and retraction of protrusion, protrusion at the cell's leading edge, attachment of trailing edge

Answer: A

Chapter Section: 14.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 14.2

Global LO: G2

24) During the sliding process in microtubules in cilia and flagella, the stalks of an axonemal dynein attach to, produce force on, and detach from a(n) _____ in a cyclic manner.

- A) A tubule
- B) B tubule
- C) radial spoke
- D) interdoubtlet nexin connections
- E) central pair

Answer: B

Chapter Section: 14.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1

Global LO: G2

25) In striated muscle contraction, the Ca^{2+} -ATPase pump

- A) releases Ca^{2+} from the T tubule.
- B) secretes Ca^{2+} from the axon terminal.
- C) is voltage-gated.
- D) decreases Ca^{2+} in the sarcoplasm.
- E) increases Ca^{2+} binding to TnC.

Answer: D

Chapter Section: 14.4

Bloom's Taxonomy: Analysis

Learning Outcome: 14.2

Global LO: G2

26) Which of the following proteins attaches a crawling cell to the substrate in a focal adhesion?

- A) actin
- B) myosins
- C) α -actinin
- D) integrin
- E) titin

Answer: D

Chapter Section: 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

27) Which of the following sequences is in the correct order for the cyclic process by which a myosin head generates force on an actin filament for muscle contraction?

- A) power stroke, crossbridge formation, crossbridge dissociation, cocking of myosin head
- B) crossbridge formation, power stroke, crossbridge dissociation, cocking of myosin head
- C) crossbridge formation, cocking of myosin head, power stroke, crossbridge dissociation
- D) crossbridge formation, crossbridge dissociation, cocking of myosin head, power stroke
- E) crossbridge formation, power stroke, crossbridge dissociation, power stroke, cocking of myosin head

Answer: B

Chapter Section: 14.4

Bloom's Taxonomy: Analysis

Learning Outcome: 14.2

Global LO: G2

28) Axonemal dyneins are

- A) associated with organelle movement.
- B) associated with cilia and flagella.
- C) monomeric myosin-like proteins.
- D) responsible for axonal anterograde transport.
- E) All of the above are correct.

Answer: B

Chapter Section: 14.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1

Global LO: G2

29) Which of the following is responsible for retrograde axonal transport?

- A) cytoplasmic dynein
- B) kinesin 5
- C) kinesin 13
- D) myosin V
- E) kinesin 1

Answer: A

Chapter Section: 14.1

Bloom's Taxonomy: Analysis

Learning Outcome: 14.1

Global LO: G2

30) In the sliding filament model of muscle contraction, I bands contain

- A) thick filaments.
- B) thin filaments.
- C) myosin II.
- D) myomesin.
- E) both thick and thin filaments.

Answer: B

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

31) Each of the following is an example of microfilament-based motility and contractility *except*

- A) ciliary motility.
- B) pseudopodia formation.
- C) cytoplasmic streaming.
- D) cytokinesis.
- E) trailing edge retraction in a moving cell.

Answer: A

Chapter Section: 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

32) Which of the following would you expect to observe in an in vitro experimental mixture containing microtubules anchored by their minus ends to centrosomes, polystyrene beads with kinesin or dynein motors attached, and ATP?

- A) movement of kinesin-attached beads along microtubules away from centrosomes
- B) movement of dynein-attached beads along microtubules toward from centrosomes
- C) movement of beads in either direction along a single microtubule
- D) removal of ATP stopping all bead movement along microtubules
- E) All would be expected.

Answer: E

Chapter Section: 14.1

Bloom's Taxonomy: Evaluation

Learning Outcome: 14.1

Global LO: G2

33) Kinesin 1 and myosin II share each of the following features *except*

- A) two heads.
- B) dimers formed by heavy chain tails coiled around each other.
- C) formation of bipolar filaments.
- D) plus end directed motor activity.
- E) ATPase activity.

Answer: C

Chapter Section: 14.1, 14.3

Bloom's Taxonomy: Analysis

Learning Outcome: 14.1, 14.2

Global LO: G2

34) In the axoneme, sidearms

- A) are 10 protofilament incomplete microtubules attached to 13 protofilament microtubules.
- B) generate force to slide microtubules past one another.
- C) project toward the central pair from each outer doublet.
- D) link adjacent doublet microtubules to limit sliding.
- E) are complete 13 protofilament microtubules.

Answer: B

Chapter Section: 14.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1

Global LO: G2

35) In the axoneme, nexin

- A) transports tubulin to the axoneme tip for assembly.
- B) generates force for outer doublets sliding past each other.
- C) projects inward from each doublet toward the central pair microtubules.
- D) links adjacent outer doublets to limit sliding.
- E) connects each outer doublet to the plasma membrane.

Answer: D

Chapter Section: 14.4

Bloom's Taxonomy: Analysis

Learning Outcome: 14.2

Global LO: G2

36) Which of the following would you use to inhibit the contraction of muscle *in vitro* as a control for an experiment you are conducting?

- A) a drug that increases sarcoplasm Ca^{2+} concentration
- B) a motor protein ATPase inhibitor
- C) a Ca^{2+} -ATPase inhibitor
- D) a drug that increases T-tubule depolarization
- E) All would inhibit muscle contraction.

Answer: B

Chapter Section: 14.4

Bloom's Taxonomy: Synthesis

Learning Outcome: 14.2

Global LO: G1

37) Kinesin 1 is highly processive motor, taking more than 100 steps (covering 1 μm along a microtubule) before dissociating, primarily because

- A) it hydrolyzes ATP more rapidly than other motors.
- B) one of its two heads is always attached in its hand-over-hand mechanism.
- C) light chains at its nonmotor end attach to cargo vesicles.
- D) its efficiency of converting ATP hydrolysis energy to useful work is 60–70%.
- E) All are required for processivity.

Answer: B

Chapter Section: 14.1

Bloom's Taxonomy: Analysis

Learning Outcome: 14.1

Global LO: G7

38) Primary cilia have a(n) _____ function and typically display a _____ structure.

- A) axonal; 9+2
- B) motile; 9+0
- C) motile; 9+2
- D) sensory; 9+0
- E) filopodial; 9+4

Answer: D

Chapter Section: 14.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1

Global LO: G7

39) A Ca^{2+} -binding protein that activates many enzymes, including myosin light-chain kinase, is

- A) Ca^{2+} -ATPase.
- B) gelsolin.
- C) calmodulin.
- D) myosin light-chain phosphatase.
- E) TnC.

Answer: C

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

40) Addition of ATP to isolated axonemes causes localized bending. What can be done to the isolated axonemes to cause the outer doublets to slide past each other unrestrained, lengthening the overall structure instead of causing bending?

- A) add gelsolin
- B) remove the interdoublet nexin links
- C) add nocodazole
- D) remove the ATP
- E) inhibit tektin activity with a drug

Answer: B

Chapter Section: 14.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 14.1

Global LO: G2

14.2 Matching Questions

Match the motor molecule to its function.

- A) processive movement toward microtubule plus end
- B) movement along microtubule toward its minus end
- C) processive movement toward actin filament plus end
- D) generates force on actin filaments in a sarcomere
- E) generates force for sliding of flagellar microtubules

1) cytoplasmic dynein

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

2) axonemal dynein

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

3) kinesin

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

4) myosin V dimer

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

5) myosin II filament

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

Answers: 1) B 2) E 3) A 4) C 5) D

Match the axoneme structure with its characteristic.

- A) links adjacent doublets
- B) generates force to slide microtubules past one another
- C) complete 13 protofilament microtubule
- D) project inward at each doublet
- E) incomplete 10–11 protofilament microtubule

6) A tubule

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

7) B tubule

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

8) sidearms

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

9) radial spokes

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

10) nexin

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

Answers: 6) C 7) E 8) B 9) D 10) A

Match the skeletal muscle protein on the left with its function on the right.

- A) binds along the length of thin filaments to regulate myosin head association
- B) links thick filaments to Z line
- C) stabilizes thin filaments
- D) major component of thin filaments
- E) bundles parallel arrays of actin filaments at Z line
- F) major component of thick filaments
- G) M line myosin filament-binding protein
- H) mediates Ca^{2+} regulation of contraction
- I) major protein of sarcoplasmic reticulum
- J) attaches actin filaments to Z line

11) actin

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

12) myosin

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

13) tropomyosin

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

14) troponin

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

15) titin

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

16) nebulin

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

17) myomesin

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

18) α -actinin

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

19) Ca²⁺-ATPase

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

20) CapZ

Chapter Section: 14.1, 14.2, 14.3, 14.4, 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1, 14.2

Global LO: G2

Answers: 11) D 12) F 13) A 14) H 15) B 16) C 17) G 18) E 19) I 20) J

Match the experimental technique with information obtained.

- A) 3D visualization of motor movement along a filament
- B) kinesin 1 step size
- C) whether a microtubule or microfilament motor protein is responsible for cell activity
- D) how much force is generated by a motor protein

21) optical laser trapping (tweezers)

Chapter Section: Key Technique

Bloom's Taxonomy: Analysis

Learning Outcome: 14.1, 14.2

Global LO: G1

22) Atomic Force Microscopy (AFM)

Chapter Section: Key Technique

Bloom's Taxonomy: Analysis

Learning Outcome: 14.1, 14.2

Global LO: G1

23) fluorescent spot tracking

Chapter Section: Key Technique

Bloom's Taxonomy: Analysis

Learning Outcome: 14.1, 14.2

Global LO: G1

24) drug inhibition of ATPase activity

Chapter Section: Key Technique

Bloom's Taxonomy: Analysis

Learning Outcome: 14.1, 14.2

Global LO: G1

Answers: 21) D 22) A 23) B 24) C

14.3 Short Answer Questions

1) Name each of the three polypeptides of troponin chains and identify what each binds to.

_____ binds to _____

_____ binds to _____

_____ binds to _____

Answer: TnT–tropomyosin; TnC–Ca²⁺; TnI–actin thin filament

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

2) When migrating cells move toward a greater concentration of a diffusible chemical, the chemical is called a _____.

Answer: chemoattractant

Chapter Section: 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

3) Cyclosis (cytoplasmic streaming) is driven by _____ that interact with _____ that are anchored to chloroplasts near an algal cell wall.

Answer: myosin motors; actin filaments

Chapter Section: 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

4) Striated muscle myofibrils are end-to-end associations of the fundamental contractile units of the muscle cell known as _____.

Answer: sarcomeres

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

5) Rigor mortis after death is caused by myosin crossbridges that fail to dissociate from the actin thin filaments because of lack of _____.

Answer: ATP

Chapter Section: 14.4

Bloom's Taxonomy: Application

Learning Outcome: 14.2

Global LO: G2

6) In the T-tubule triad structure seen in electron micrographs of striated muscle, the T tubule is flanked by two terminal cisternae of the _____. This association facilitates opening terminal cisternae ryanodine receptors, which _____ to initiate sarcomere contraction.

Answer: sarcoplasmic reticulum; release Ca^{2+} from the SR into the sarcoplasm

Chapter Section: 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

7) Crawling cells produce broad, thin protrusions known as _____ and pointed structures called _____ at the leading edge of the cell.

Answer: lamellipodia; filopodia

Chapter Section: 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

8) The contractile _____ are aligned in _____ and _____ striated muscles but not in _____ muscles.

Answer: sarcomeres/myofibrils; cardiac; skeletal; smooth

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

9) _____ is a directional movement in response to a graded chemical stimulus.

Answer: Chemotaxis

Chapter Section: 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

10) Myosin II is ideally suited for muscle contraction because it forms _____ filaments.

Answer: bipolar

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

11) Microtubule motors _____ move vesicles toward the minus end of a microtubule.

Answer: cytoplasmic dynein

Chapter Section: 14.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1

Global LO: G2

12) The _____ protein complex facilitates the interaction between cytoplasmic dynein, a cargo vesicle, and a microtubule.

Answer: dynactin

Chapter Section: 14.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1

Global LO: G2

13) The _____ element of the sarcoplasmic reticulum is where _____ function to decrease sarcoplasmic Ca^{2+} concentration for muscle relaxation.

Answer: medial; Ca^{2+} -ATPases

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

14) _____ is the process that moves tubulin subunits to the flagellar tip where they assemble onto axonemal microtubules.

Answer: Intraflagellar transport

Chapter Section: 14.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.1

Global LO: G2

15) Mutations in myosin V cause a human disorder known as _____, involving partial albinism and neurological defects.

Answer: Griscelli's disease

Chapter Section: 14.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

14.4 Inquiry

1) The structural organization of muscle fibers makes it possible to produce contractile force.

a. What is the fundamental contractile unit of skeletal muscle?

b. What are the names of the bands, lines and zone of the contractile unit of skeletal muscle?

c. Each of the contractile unit region names in (b) contains a letter. What does each of these letters stand for?

d. What changes occur in the contractile unit regions during contraction?

Answer:

a. The sarcomere is the fundamental contractile unit of the muscle cells.

b. A and I bands, M and Z lines, H zone

c. A–anisotropic (with respect to plane-polarized light), I–isotropic (with respect to plane-polarized light), M–contains myomesin, Z–*zwischen* (meaning between), H–*hell* (meaning light)

d. The A band and M line remain unchanged, but the distance between Z lines gets shorter as the I band and H zone get shorter or disappear.

Chapter Section: 14.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

2) Eukaryotic motile cilia and flagella have similar structures but vary in number, length, and function. For each of the following, compare/contrast cilia and flagella.

- a. number present per cell
- b. length of appendage
- c. direction of force generated relative to the direction of the appendage
- d. function in unicellular and multicellular eukaryotes
- e. motion analogy

Answer:

- a. Flagellated cells have one or a few flagella. Ciliated cells have numerous cilia.
- b. Flagella generally are longer than cilia.
- c. Flagella generate a force parallel to the flagellum axis, whereas cilia generate a force perpendicular to the cilium axis.
- d. Flagella move cells in both multicellular and unicellular eukaryotes. Cilia move unicellular eukaryotes and small multicellular organisms, but ciliated cells in larger multicellular organisms use cilia to move the environment rather than the body as a whole.
- e. The motion of flagella is symmetrical, undulatory, and can be helical. Cilia display an oarlike pattern of beating, with a power and a recovery stroke.

Chapter Section: 14.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 14.2

Global LO: G2

3) The tetanus toxin tetanospasmin blocks release of the neurotransmitter (GABA) that inhibits motor neurons and decreases muscle contraction. This causes persistence of the signal to contract and the resulting violent muscle contraction known as tetanic spasm. Compare the sustained contraction of muscle cells in rigor mortis with muscle contraction caused by the action of tetanospasmin on motor neurons.

Answer: In the case of rigor mortis, the ATP supply is depleted because the cells are in various stages of dying. Loss of ATP causes myosin head crossbridges to lock onto actin thin filaments. The cells accumulate crossbridges over time until ATP is depleted from the cell.

In the case of tetanospasmin, the continuous stimulation of contraction prevents lowering of sarcoplasmic Ca^{2+} ion concentration by stimulating continual release of Ca^{2+} pumped back into the sarcoplasmic reticulum by Ca^{2+} -ATPases. As a result, the consistent activating levels of Ca^{2+} in cytosol keep the muscle cells in a chronic state of contraction.

Chapter Section: 14.4

Bloom's Taxonomy: Synthesis

Learning Outcome: 14.2

Global LO: G7, G8

4) You identify a protein that may play a role in microtubule-based transport. Propose a series of experiments that would aid you in confirming your assessment that this protein has a role in microtubule-based transport.

Answer: Possible approaches include: (1) deleting the gene from cells and looking for a disruption of microtubule-based transport; (2) treating cells with potential inhibitors to inhibit cellular activity and restoring activity by removal of the inhibitor; (3) linking association of the gene deficiency with a genetic disease characterized by defective microtubule-based transport.

Chapter Section: 14.1

Bloom's Taxonomy: Synthesis

Learning Outcome: 14.1

Global LO: G1

5) Consider the observations that a label on one head of a myosin V dimer moves in steps of 72 nm, but a label much nearer the myosin tail moves in steps of 36 nm. Discuss how these observations support a hand-over-hand mechanism rather than an inchworm mechanism of myosin V motility on an actin filament.

Answer: In the inchworm model, each myosin head would move in steps of 36 nm, as the trailing head moves 36 nm up to the position just behind the leading head and the leading head then steps forward 36 nm. In the hand-over-hand model, the trailing head steps a total of 76 nm as it pivots past the leading head to become the new leading head, whereas the tail region moves only 36 nm forward (see Fig. 14A-2).

Chapter Section: Key Technique

Bloom's Taxonomy: Analysis

Learning Outcome: 14.2

Global LO: G1

6) Explain why myosin II in a sarcomere can produce force only for that sarcomere contraction and not for that sarcomere extension.

Answer: In each half sarcomere, myosin heads on each end of a bipolar filament can step only toward the actin thin filament plus end, which is anchored to the Z line on that end of the sarcomere. This causes contraction of the sarcomere. To generate force for sarcomere extension, myosin II heads would need to produce force by stepping toward the minus end of the actin filament, which only myosin VI (of the myosins described in the text) is capable of.

Chapter Section: 14.4

Bloom's Taxonomy: Evaluation

Learning Outcome: 14.2

Global LO: G2, G8

7) Explain why sarcomere tension increases as the sarcomere length shortens from an extended length of 3.5 μm to 2 μm but then decreases if the sarcomere contracts to a length of 1.5 μm .

Answer: At the extended length, some myosin heads on each end of the bipolar filaments are not overlapping with actin thin filaments and therefore cannot generate force. As the sarcomere shortens, more of the myosin heads overlap with thin filaments and add to the force generated. At a sarcomere length of 2 μm , all of the myosin heads are positioned to interact with the thin filaments and maximal force is generated. Further sarcomere shortening causes the ends of actin thin filaments from one half sarcomere to interfere with myosin crossbridge formation in the other half sarcomere, decreasing the number of force-producing crossbridges and sarcomere tension.

Chapter Section: 14.4

Bloom's Taxonomy: Synthesis

Learning Outcome: 14.2

Global LO: G2, G8

Becker's The World of the Cell, 9e (Hardin/Bertoni/Kleinsmith)

Chapter 15 Beyond the Cell: Cell Adhesions, Cell Junctions, and Extracellular Structures

15.1 Multiple-Choice Questions

1) Which of the following stabilizes homophilic interactions between the extracellular domains of cadherins on neighboring cells?

- A) hyaluronate
- B) proteoglycans
- C) Ca^{2+}
- D) integrins
- E) talin

Answer: C

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

2) Polarized cells are characterized by

- A) random orientation within a tissue.
- B) being surrounded on all sides by a basement membrane.
- C) junctions only to ECM and not to other cells.
- D) distinct apical and basolateral ends of the cell.
- E) formation of several connected layers of cells.

Answer: D

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

3) Each of the following cell junctions is found in intestinal epithelial cells *except*

- A) gap junctions.
- B) hemidesmosomes.
- C) tight junctions.
- D) desmosomes.
- E) All are found in intestinal epithelial cells.

Answer: E

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

4) In animals, the extracellular matrix plays important roles in each of the following processes *except*

- A) cell division.
- B) cell motility.
- C) DNA replication.
- D) cell differentiation.
- E) cell adhesion.

Answer: C

Chapter Section: 15.0

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

5) An autoimmune disorder in which an immune response targets Type V collagen fibers is most likely to affect the

- A) skin.
- B) heart.
- C) lungs.
- D) stomach.
- E) tongue.

Answer: A

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

6) Which of the following junctions is most abundant in tissues such as skin that are required to resist significant mechanical stress?

- A) gap junctions
- B) desmosomes
- C) focal adhesions
- D) tight junctions
- E) All are abundant in tissues resisting stress.

Answer: B

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

7) The primary function of each of the following is for adhesion of a cell to another cell or the extracellular matrix *except*

- A) desmosomes.
- B) hemidesmosomes.
- C) plaques.
- D) connexins.
- E) cadherins.

Answer: D

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

8) Mice lacking plakoglobin have skin defects but usually die from

- A) cancer.
- B) muscular dystrophy.
- C) heart failure.
- D) Ehlers–Danlos syndrome.
- E) scurvy.

Answer: C

Chapter Section: 15.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 15.1

Global LO: G7

9) N-CAM mediates interactions between neural cells and is a member of which of the following protein families?

- A) catenins
- B) cadherins
- C) selectins
- D) immunoglobulin superfamily (IgSF) proteins
- E) lectins

Answer: D

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

10) Selectins mediate interactions between which of the following?

- A) leukocytes and blood vessel endothelial cells
- B) muscle cells and ECM
- C) nerve cells and other nerve cells
- D) intestinal epithelial cells with neighboring cells
- E) skin cells in different skin layers

Answer: A

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

11) Each of the following components is associated with the primary cell wall *except*

- A) cellulose microfibrils.
- B) hemicelluloses.
- C) pectins.
- D) glycoproteins.
- E) lignins.

Answer: E

Chapter Section: 15.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

12) Which of the following is the correct order for the synthesis of the plant cell wall?

- A) middle lamella; primary cell wall; secondary cell wall
- B) primary cell wall; middle lamella; secondary cell wall
- C) secondary cell wall; primary cell wall; middle lamella
- D) middle lamella; secondary cell wall; primary cell wall
- E) primary cell wall; secondary cell wall; middle lamella

Answer: A

Chapter Section: 15.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

13) You discover a dog suffering from an autoimmune response. On further investigation, you find that the cells of various tissues are loosely arranged and do not have characteristic shapes, and the animal is prone to hemorrhaging because of poor blood clotting. Based on these results, the target of the autoimmune response is most likely to be

- A) fibronectins.
- B) laminins.
- C) extensins.
- D) pectins.
- E) elastins.

Answer: A

Chapter Section: 15.2

Bloom's Taxonomy: Application

Learning Outcome: 15.2

Global LO: G7

14) The finding that integrin-linked kinase (ILK) is required for focal adhesion function but its kinase activity is not, suggests the protein acts as a

- A) direct connection to the ECM.
- B) linker protein.
- C) scaffolding protein.
- D) cytoskeletal filament.
- E) signaling protein.

Answer: C

Chapter Section: 15.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 15.1

Global LO: G2

15) Each of the following is a component of focal adhesions *except*

- A) integrin.
- B) talin.
- C) α -actinin.
- D) vinculin.
- E) plakoglobin.

Answer: E

Chapter Section: 15.1

Bloom's Taxonomy: Application

Learning Outcome: 15.1

Global LO: G2

16) Which of the following functions like a fence to block membrane protein movement between the apical and basolateral regions of an intestinal epithelial cell plasma membrane?

- A) gap junctions
- B) desmosomes
- C) adherens junction
- D) hemidesmosomes
- E) tight junction

Answer: E

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

17) Ehlers–Danlos syndrome and scurvy both are characterized by defects in

- A) fibronectin.
- B) desmosomes.
- C) collagen.
- D) dystrophin.
- E) elastin.

Answer: C

Chapter Section: 15.2

Bloom's Taxonomy: Application

Learning Outcome: 15.2

Global LO: G7

18) Which of the following pairs shares a similar function?

- A) adhesive junction–tight junction
- B) desmotubule–gap junction
- C) tight junction–gap junction
- D) plaque–plasmodesma
- E) connexon–desmosome

Answer: B

Chapter Section: 15.1

Bloom's Taxonomy: Application

Learning Outcome: 15.2

Global LO: G7

19) Which of the following regulates paracellular transport?

- A) adherens junctions
- B) gap junctions
- C) desmosomes
- D) tight junctions
- E) hemidesmosomes

Answer: D

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

20) While investigating a medical condition, you discover that the patient has an autoimmune disease that targets and destroys fibroblasts. This disorder would most likely affect

- A) fibrin activity in blood clotting.
- B) annulus formation.
- C) hyaluronate lubrication of joints.
- D) collagen production.
- E) primary cell wall deposition.

Answer: D

Chapter Section: 15.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 15.2

Global LO: G2

21) Which of the following is associated with the lubrication of joints?

- A) chondroitin sulfate
- B) lignins
- C) hyaluronate
- D) keratan sulfate
- E) pectins

Answer: C

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

22) Which of the following aspects of collagen would be directly affected by a defect in procollagen peptidase?

- A) synthesis
- B) secretion
- C) extracellular assembly
- D) binding to fibronectin
- E) all of the above

Answer: C

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

23) Which of the following could you use experimentally to help you determine which carbohydrates are present on a cell's surface?

- A) cadherins
- B) proteoglycans
- C) pectins
- D) connexins
- E) lectins

Answer: E

Chapter Section: 15.1

Bloom's Taxonomy: Application

Learning Outcome: 15.1

Global LO: G1

24) Each of the following is a good functional match between animal and plant extracellular structures or fibers *except*

- A) elastins–extensins.
- B) extracellular matrix–cell wall.
- C) collagens–cellulose.
- D) proteoglycans–hemicelluloses.
- E) fibronectins–pectins.

Answer: A

Chapter Section: 15.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 15.2

Global LO: G7

- 25) Elastin molecules are crosslinked to each other by
- A) hydrogen bonds between hydroxylysines and hydroxyprolines.
 - B) covalent bonds between lysines.
 - C) covalent bonds between prolines.
 - D) hydrophobic interactions between hydroxyprolines and glycines.
 - E) peptide bonds between glycines.

Answer: B

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

- 26) Each of the following is true about basal lamina *except*
- A) it serves as a structural support.
 - B) it allows for movement of some small molecules but blocks movement of large molecules.
 - C) it can influence cell migrations in certain areas of the body.
 - D) it contains collagen type IV, proteoglycans, laminins, and entactin.
 - E) it is composed primarily of fibronectin.

Answer: E

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

- 27) You suspect that an elderly patient's skin blistering is caused by an autoimmune disorder. If you are correct, the patient's serum likely contains antibodies directed against

- A) talin.
- B) vinculin.
- C) α -actinin.
- D) BPAG2.
- E) actin.

Answer: D

Chapter Section: 15.2

Bloom's Taxonomy: Application

Learning Outcome: 15.2

Global LO: G7

28) To investigate the role of cadherin in early development, you continuously treat frog zygotes with antibodies that bind EP-cadherin. You most likely would find that

- A) cell division stops but the existing cells organize to form a small, properly organized blastula.
- B) the cells associate only loosely and fail to form a properly organized blastula.
- C) the cells associate only loosely but skip the blastula stage to develop into a normal frog.
- D) the cells divide uncontrollably and form a large tumor-like mass.
- E) blastula formation and subsequent development proceed normally.

Answer: B

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

29) Collagens interact directly with which of the following types of junctions?

- A) gap junctions
- B) tight junctions
- C) hemidesmosomes
- D) desmosomes
- E) None of the above are correct.

Answer: E

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

30) The primary function of focal contacts is

- A) cell–ECM adhesion.
- B) cell–cell adhesion.
- C) cell–cell communication.
- D) paracellular pathway regulation.
- E) establishing cell polarity.

Answer: A

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G7

31) Which of the following is responsible for the impermeability of the epithelium lining the bladder and ducts connecting the pancreas to the digestive tract?

- A) adherens junctions
- B) gap junctions
- C) desmosomes
- D) tight junctions
- E) hemidesmosomes

Answer: D

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

32) The type of adhesive junction that connects epithelial cells to the basal lamina is a

- A) hemidesmosome.
- B) desmosome.
- C) gap junction.
- D) tight junction.
- E) focal contact.

Answer: A

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

33) Which of the following forms the insoluble layer of aromatic alcohols found primarily in woody tissues?

- A) hyaluronates
- B) cadherins
- C) proteoglycans
- D) lignins
- E) pectins

Answer: D

Chapter Section: 15.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

34) Animal extracellular matrices contain each of the following *except*

- A) collagens.
- B) proteoglycans.
- C) pectins.
- D) fibronectins.
- E) elastins.

Answer: C

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

35) In cartilage, proteoglycans form large complexes by attaching to

- A) glycosaminoglycans.
- B) collagen.
- C) hyaluronate.
- D) laminin.
- E) elastin.

Answer: C

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

36) Costameres, in which cytosolic actin filaments are linked to extracellular laminins through dystrophin–dystroglycan complexes, are found in

- A) skin cells.
- B) intestinal epithelial cells.
- C) striated muscle cells.
- D) nerve cells.
- E) pancreatic cells.

Answer: C

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

37) Each of the following is a component of the plant cell wall network *except*

- A) hyaluronate.
- B) cellulose.
- C) extensin.
- D) hemicellulose.
- E) pectin.

Answer: A

Chapter Section: 15.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

38) Which of the following is a primary structural constituent of the plant cell secondary wall?

- A) hemicellulose
- B) extensin
- C) collagen
- D) pectin
- E) cellulose

Answer: E

Chapter Section: 15.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

39) A cluster of intracellular linker proteins associated with a junction is called a(n)

- A) annulus.
- B) connexon.
- C) plaque.
- D) tonofilament.
- E) accessory protein complex.

Answer: C

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

40) Fibronectin interacts with each of the following *except*

- A) fibrin.
- B) collagen.
- C) heparin.
- D) integrin.
- E) pectin.

Answer: E

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

41) Although MCF-10A breast cancer cells grow well on Matrigel-coated culture dishes, the

cells will form cysts of polarized cells that produce milk only when

- A) extra collagen is added to the Matrigel.
- B) extra fibronectin is added to the Matrigel.
- C) extra elastin is added to the Matrigel.
- D) the Matrigel layer is thick enough to form a 3D environment.
- E) the Matrigel layer is present as a very thin layer on the culture dishes.

Answer: D

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

42) Each of the following proteins is a component of both costameres and focal adhesions *except*

- A) β 1-integrin.
- B) vinculin.
- C) talin.
- D) α -actinin.
- E) dystrophin.

Answer: E

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

43) Each of the following is true about plant pectins *except*

- A) pectins are long unbranched polymers.
- B) pectins form the matrix in which cellulose microfibrils are embedded.
- C) pectins bind adjacent cell walls together.
- D) pectins trap and bind water molecules.
- E) pectins have a gel-like consistency.

Answer: A

Chapter Section: 15.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

- 44) Each of the following is true about plant rosettes *except*
- A) rosettes are localized in the plasma membrane.
 - B) rosettes are cellulose-synthesizing enzyme complexes.
 - C) rosette deposition of cellulose microfibrils is directed by microtubules.
 - D) rosettes move in the plasma membrane to deposit growing cellulose microfibrils.
 - E) rosettes deposit cellulose microfibrils only in the middle lamella.

Answer: E

Chapter Section: 15.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

- 45) Expansins are proteins that loosen up cell walls so cells can elongate. Auxin is thought to activate expansins by

- A) aligning cellulose microfibrils.
- B) activating protons pumps to decrease pH in a section of the cell wall.
- C) depositing lignin molecules between cellulose fibrils.
- D) increasing turgor pressure.
- E) increasing pectin hydration.

Answer: B

Chapter Section: 15.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

15.2 Matching Questions

Match the cell adhesion or junction on the left with the appropriate choice on the right.

- A) junction connecting tonofilaments to collagen filaments
- B) anchors cytoskeleton of one cell with that of a neighboring cell
- C) type of adherens junction connecting actin filaments to the extracellular matrix
- D) dense cluster of intracellular linker proteins connected to cytoskeletal filaments
- E) protein subunit of gap junctions
- F) continuous attachment zones connecting actin microfilament structures near the apical end of epithelial cells
- G) button-like point of strong adhesion between two cells
- H) intermediate filaments associated with plaques
- I) create a permeability barrier between epithelial cells
- J) an adhesive junction that connects an epithelial cell to the basal lamina
- K) hollow cylinders associated with gap junctions
- L) connects cytoplasms of two cells
- M) specialized modifications of the cell membrane in which two cells come together in multicellular organisms

1) adherens junctions

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

2) desmosome

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

3) cell junctions

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

4) connexons

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

5) plaque

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

6) tonofilaments

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

7) hemidesmosome

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

8) adhesive junction

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

9) tight junctions

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

10) connexin

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

11) gap junction

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

12) focal adhesion

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

Answers: 1) F 2) G 3) M 4) K 5) D 6) H 7) J 8) B 9) I 10) E 11) L 12) C

15.3 Short Answer Questions

1) The _____ is the layer of cell wall material shared between adjacent plant cells.

Answer: middle lamella

Chapter Section: 15.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

2) _____ are cytosolic channels through openings in plant cell walls.

Answer: Plasmodesmata

Chapter Section: 15.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

3) _____ are insoluble polymers of aromatic alcohols that primarily are found in woody tissues.

Answer: Lignins

Chapter Section: 15.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

4) A(n) _____ is the ring of cytoplasm through a plasmodesmata.

Answer: annulus

Chapter Section: 15.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

5) _____ are Ca^{2+} -dependent adhesive glycoproteins that function in animal cell-cell recognition and adhesion.

Answer: Cadherins

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

6) A cadherin forms a _____ extracellular interaction with a _____ on another cell.

Answer: homophilic; cadherin

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

7) Animal cells attach to adhesive glycoproteins such as _____ and _____ in the extracellular matrix.

Answer: fibronectin; laminin

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

8) Numerous collagen fibrils associate to form a collagen _____.

Answer: fiber

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

9) _____ are polysaccharides composed of repeating disaccharide units that are the carbohydrate part of some proteoglycans.

Answer: Glycosaminoglycans (GAGs)

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

10) One of the most important roles of proteoglycans is to trap _____ to form a _____ matrix.

Answer: water; gelatinous

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

11) Free _____ molecules provide lubrication to joints between movable bones.

Answer: hyaluronate

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

12) The _____ serves as structural support in tissue organization and as a permeability barrier.

Answer: basal lamina

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

13) _____ are the animal cell surface proteins that link the cytoskeleton to the extracellular matrix.

Answer: Integrins

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G2

14) The three most common glycosaminoglycans (GAGs) are _____, _____, and _____.

Answer: chondroitin sulfate; keratan sulfate; hyaluronate

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

15) A major component of collagen is the amino acid _____, which is the only one small enough to fit in the interior of the collagen triple helix structure.

Answer: glycine

Chapter Section: 15.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.2

Global LO: G2

15.4 Inquiry

1) You are investigating the activity of different cadherins in cells. You use recombinant DNA technology to express GFP-E-cadherin (green) in some L-cells (which do not normally contain cadherin) and YFP-P-cadherin (yellow) in other L-cells. Explain results you expect in three cultures containing: (1) only cells expressing GFP-E-cadherin, (2) only cells expressing YFP-P-cadherin, and (3) a mixture of GFP-E-cadherin-expressing and YFP-P-cadherin-expressing cells. Also predict how (a) knocking down the ability of the cells to synthesize fibronectin (FN) or (b) adding EGTA, which depletes free Ca^{2+} concentration in the culture medium, might affect the results.

Answer:

- 1) The GFP-E-cadherin-expressing cells would form adherens junctions (labeled green) with one or more cells, possibly forming a sheet of cells.
- 2) The YFP-E-cadherin-expressing cells would form adherens junctions (labeled yellow) with one or more cells, possibly forming a sheet of cells.
- 3) Both cell types would form adherens junctions but only with cells expressing the same type of cadherin, because cadherin extracellular interactions are cadherin-type specific.
 - a) Cells expressing either labeled cadherin would be expected to form adherens junctions regardless of whether they can synthesize FN or not, but they may not adhere and spread on the substrate as well without FN.
 - b) Medium lacking Ca^{2+} will not support the formation of cadherin-dependent adherens junctions, because cadherin–cadherin interaction depends on the presence of extracellular Ca^{2+} .

Chapter Section: 15.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 15.1

Global LO: G1, G7, G8

2) Describe experiments using marker molecules that have been done to elucidate important properties of (1) tight junctions and (2) gap junctions, and propose one additional experiment you could do with each junction to gain a greater understanding of that junction's properties.

Answer:

1) When an electron-opaque tracer was added to the basolateral side of an epithelial layer, the tracer penetrated in the intracellular space between the cells, past desmosomes and the adherens junction, up to the position of the tight junction. The tracer did not pass through the tight junction. This demonstrated the barrier property of the tight junction in regulating paracellular transport, at least from the basolateral side to the apical side. A possible additional experiment would be to add the tracer to the apical side of the epithelium to assess whether the tight junction also blocks paracellular transport from that direction.

2) Microinjection of various sizes of fluorescent molecules into one cell connected to another by gap junctions revealed that gap junctions pass molecules between the cytoplasm of cells that are up to a size limitation of ~1200 Daltons MW. A possible additional experiment would be to test molecules of different shapes, say long linear molecules that are >1200 D, or molecules with different chemical characteristics to determine whether size is the only limitation. Another possible experiment would be to determine what happens to gap junctions when one of the cells dies, perhaps by loading one cell with a small fluorescent probe and then puncturing the other cell to determine whether the dye leaks from the living cell gap junctions or whether the gap junctions can close.

Chapter Section: 15.1

Bloom's Taxonomy: Synthesis

Learning Outcome: 15.1

Global LO: G1, G7, G8

3) Discuss the role of claudins in the structure and functions of tight junctions. Propose what might change for a tight junction in which charged amino acids in the claudin large extracellular loop are mutated to nonpolar amino acids.

Answer: Claudin extracellular loop domains form homophilic interactions that are a barrier to paracellular transport through intercellular spaces between epithelial cells. Charged amino acids in the claudin large extracellular loop form ion-selective pores that allow the passage of specific ions through the barrier. Mutation of the charged amino acids to nonpolar ones would create a more hydrophobic region that could block ion transport but allow passage of nonpolar substances that fail to pass through wild-type claudins. This would change the barrier function of the tissue and could affect the health of the organism. Recessive mutations in one type of claudin result in an inherited disorder characterized by severe Mg^{2+} and Ca^{2+} imbalance.

Chapter Section: 15.1

Bloom's Taxonomy: Application

Learning Outcome: 15.1

Global LO: G1, G8

4) Amniocentesis and genetic screening reveal that a fetus carries a defective *LI-CAM* gene. What defects do you expect, and why?

Answer: During embryogenesis *N-CAM* and *LI-CAM* function in the outgrowth and the bundling of neurons. Individuals expressing a defective *LI-CAM* will have defects in the brain corpus callosum, causing mental retardation and other neural defects.

Chapter Section: 15.1

Bloom's Taxonomy: Application

Learning Outcome: 15.1

Global LO: G1, G8

5) Cancer cells exhibit a variety of differences from normal cells in cell adhesion, cell junctions, and cell effects on the extracellular matrix, some of which facilitate metastasis. Explain three changes found in some cancer cells related to these aspects of cell structure and function.

Answer: Answers may include the following:

1) Cells may stop producing E-cadherin or other cadherins, which may be associated with an epithelial–mesenchymal transition and subsequent metastasis.

2) Cells may increase MMP secretion, which can cause localized ECM breakdown and facilitate metastasis through the ECM.

3) Cells may increase binding to the basal lamina, which can facilitate metastasis cell motility.

4) Cells may decrease synthesis and secretion of fibronectins, resulting in loss of cell shape and detachment from the ECM.

5) Cells may develop focal adhesion unresponsiveness to the Wnt signaling pathway, which normally would repress cell division and motility.

6) Cells may lose anchorage-dependence for growth, possibly through permanent FAK activation.

Chapter Section: 15.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 15.1, 15.2

Global LO: G1, G8

23.1 Multiple-Choice Questions

1) Inhibitors of which of the following would interfere with nitric oxide stimulation of cardiac artery dilation through relaxation of smooth muscle cells?

- A) protein kinase G.
- B) nitric oxide synthase
- C) guanylyl cyclase
- D) calmodulin
- E) Inhibitors of all of the above would interfere with nitric oxide stimulation.

Answer: E

Chapter Section: 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

2) The concentration of the primary messenger (ligand) in the bloodstream changes from $<10^{-9}$ M when the signal is not being secreted to 10^{-7} M when it is being secreted. Which of the following K_d s for binding that ligand would be optimal for a receptor involved in a responsive signaling system?

- A) 10^{-10}
- B) 10^{-9}
- C) 10^{-8}
- D) 10^{-7}
- E) 10^{-6}

Answer: C

Chapter Section: 23.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G4

3) Each of the following is an intracellular messenger *except*

- A) calcium ions.
- B) IP₃.
- C) DAG.
- D) cAMP.
- E) PDGF.

Answer: E

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

- 4) In a hormone-stimulated signaling system, the concentration of hormone exposed to the receptor is
- A) always equal to the receptor K_d for binding the hormone.
 - B) never equal to the receptor K_d for binding the hormone.
 - C) below the receptor K_d for binding the hormone before stimulation of hormone secretion.
 - D) above the receptor K_d for binding the hormone before stimulation of hormone secretion.
 - E) always constant.

Answer: C

Chapter Section: 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.2

Global LO: G2

- 5) Which of the following is a lipophilic primary message that crosses the plasma membrane and interacts with a cytosolic receptor?

- A) estrogen
- B) EGF
- C) pertussis toxin
- D) $\text{TGF}\beta$
- E) PDGF

Answer: A

Chapter Section: 23.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

- 6) Desensitization of a cell to a ligand may involve each of the following *except*

- A) destruction of the ligand.
- B) decrease in receptor ability to activate intracellular signaling pathway.
- C) receptor destruction.
- D) receptor endocytosis.
- E) decrease in receptor affinity for the ligand.

Answer: A

Chapter Section: 23.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

7) Which of the following would be correctly described as a juxtacrine signaling?

- A) a signal that requires physical contact between cell
- B) a signal that acts on the cell that produces it
- C) a signal that diffuses over long distances from the producing cell
- D) a signal that diffuses over short distances from the producing cell
- E) a hormonal signal

Answer: A

Chapter Section: 23.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

8) The interleukin-2 (IL-2) growth factor interacts with receptors on

- A) embryonic cells.
- B) cytotoxic T lymphocytes.
- C) mesenchymal cells.
- D) smooth muscle cells.
- E) epithelial cells.

Answer: B

Chapter Section: 23.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.2

Global LO: G2

9) Each of the following ligands binds to an intracellular receptor *except*

- A) cortisol.
- B) retinoic acid.
- C) thyroid hormone.
- D) epinephrine.
- E) estrogen.

Answer: D

Chapter Section: 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.2

Global LO: G2

10) Each of the following activates a receptor tyrosine kinase *except*

- A) EGF.
- B) PDGF.
- C) TGF β
- D) FGF.
- E) Insulin.

Answer: C

Chapter Section: 23.3

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

Base your answers to Questions 11–15 on the following information:

You have isolated a primary signal factor from the blood of a space alien, which you name XFF (for X-files factor). You also have isolated various types of alien cells which appear to interact with XFF. Based on preliminary data, you think that XFF could be similar to one of three compounds normally found in human blood: epidermal growth factor (EGF), epinephrine, or estrogen.

11) You isolate a crude fraction from the alien cells containing receptors for XFF. You perform a binding assay and plot the XFF receptors occupied (y -axis) at several XFF concentrations. You obtain a Michaelis–Menten-like simple binding curve that indicates a 1 XFF:1 receptor interaction and approaches a maximal level of binding you define as B_{max} . From the curve you extrapolate the K_d for XFF binding as the concentration at which 50% of the receptors are bound. What value reflects the affinity of the receptor for the ligand?

- A) K_d
- B) B_{max}
- C) $K_d - B_{max}$
- D) $1/B_{max}$
- E) The affinity cannot be determined from the information available.

Answer: A

Chapter Section: 23.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G4

12) To test whether the XFF signal system is similar to that of epinephrine, you microinject a nonhydrolyzable analogue of GTP into an XFF-responsive cell. If the XFF mechanism is similar to that of epinephrine, you would expect the cellular response to be

- A) blocked completely.
- B) prolonged.
- C) delayed.
- D) 50 percent of normal.
- E) It is not possible to predict without knowing how many receptors are present on the cell.

Answer: B

Chapter Section: 23.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

13) You add a large amount of XFF to cultured alien liver cells. Which of the following would confirm that XFF is the alien equivalent of human epinephrine?

- A) increased cell division
- B) decreased cell division
- C) glycogen breakdown
- D) increased glucose uptake
- E) insensitivity to insulin

Answer: C

Chapter Section: 23.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

14) Which of the following would best be used to demonstrate XFF is most like EGF?

- A) exposure of cells to radioactive XFF
- B) receptor binding assays
- C) microinjection into cells of nonhydrolyzable GTP analogues
- D) exposure of cells to ryanodine
- E) exposure of cells to tyrosine kinase inhibitors

Answer: E

Chapter Section: 23.3

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G7

15) Previous experiments have not conclusively demonstrated XFF is the alien epinephrine, so you decide to investigate other XFF properties using radioactive XFF. Which of the following XFF behaviors confirm that XFF is similar to estrogen?

- A) XFF binds only to the cell surface.
- B) XFF diffuses into the cell cytoplasm.
- C) XFF is internalized by receptor-mediated endocytosis.
- D) Ryanodine blocks the interaction of XFF with cells.
- E) Opioids block the interaction of XFF with cells.

Answer: B

Chapter Section: 23.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.2

Global LO: G2

16) Compounds that inhibit receptors by preventing the natural messenger from binding are known as

- A) agonists.
- B) arrestins.
- C) sildenafil.
- D) syndecans.
- E) antagonists.

Answer: E

Chapter Section: 23.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

17) Each of the following is a step in the epinephrine–cAMP pathway in which the signal is amplified *except*

- A) receptor activation of G-protein.
- B) G-protein activation of adenylyl cyclase.
- C) adenylyl cyclase production of cAMP.
- D) protein kinase A phosphorylation of target proteins.

Answer: B

Chapter Section: 23.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

18) Each of the following is a G-protein coupled receptor (GPCR) or type of GPCR *except*

- A) steroid hormone receptors.
- B) opioid receptors.
- C) thyroid-stimulating hormone receptors.
- D) olfactory receptors.
- E) β -adrenergic receptors.

Answer: A

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

19) Each of the following is correct about cholera and its effect on G-protein–adenylyl cyclase activity *except*

- A) *Vibrio cholera* bacteria colonize the gut.
- B) Cholera toxin is secreted by *Vibrio cholera* bacteria.
- C) Intestinal cells infected with cholera toxin import abnormally high levels of salts and fluids into the bloodstream.
- D) Cholera toxin alters Gs α so that it no longer hydrolyzes GTP.
- E) Intestinal cells infected with cholera toxin maintain abnormally high cAMP levels.

Answer: C

Chapter Section: 23.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

20) Reception of a signal released by the same cell is which of the following types of signaling?

- A) paracrine
- B) endocrine
- C) autocrine
- D) juxtacrine
- E) exocrine

Answer: C

Chapter Section: 23.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

21) You are seeking a new anti-inflammatory drug to inhibit local responses to stress and injury. Which of the following activities would you look for in a potential anti-inflammatory agent?

- A) stimulates release of prostaglandins
- B) blocks adenylate cyclase activity
- C) blocks an enzyme associated with arachidonic acid production
- D) stimulates the activation of platelets
- E) increases estrogen secretion

Answer: C

Chapter Section: 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.2

Global LO: G2

22) Kinases and phosphatases are essential in the cell because they directly

- A) turn proteins "on and off" through changes in phosphorylation status.
- B) bind hydrophobic hormones in the cytoplasm.
- C) bind and release calcium ions.
- D) destroy IP₃, turning off that signaling pathway.
- E) activate and repress genes.

Answer: A

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

23) cAMP activates

- A) protein kinase A.
- B) protein kinase B.
- C) protein kinase C.
- D) protein kinase G.
- E) MAPK.

Answer: A

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.2

Global LO: G2

24) Which of the following is the correct order of steps in the EGF–cAMP signaling pathway?

- A) ligand binding–G-protein activation–adenylyl cyclase activation–cAMP production–protein kinase A activation
- B) ligand binding–protein kinase A activation–G-protein activation–adenylyl cyclase activation–cAMP production
- C) G-protein activation–ligand binding–adenylyl cyclase activation–cAMP production–protein kinase A activation.
- D) ligand binding–G-protein activation–adenylyl cyclase activation–cAMP production–protein kinase A activation
- E) ligand binding–adenylyl cyclase activation–G-protein activation–cAMP production–protein kinase A activation

Answer: D

Chapter Section: 23.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

25) Each of the following is a correct cell-specific response of Epinephrine–cAMP signaling *except*

- A) liver cell glycogen breakdown to glucose.
- B) stronger heart muscle contraction.
- C) stronger smooth muscle contraction.
- D) inhibition of platelet activation during blood clotting.
- E) increase in intestinal epithelial cell fluid secretion.

Answer: C

Chapter Section: 23.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

26) cAMP affects cell activities by interacting directly with

- A) adenylyl cyclase.
- B) protein kinase A regulatory subunits.
- C) protein kinase A catalytic subunits.
- D) cAMP phosphodiesterase.
- E) calmodulin.

Answer: B

Chapter Section: 23.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

27) The low level of cytosolic calcium ions in unactivated cells is established by which of the following?

- A) calmodulin
- B) calcium ATPases
- C) kinases
- D) phosphatases
- E) phospholipase C

Answer: B

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.3

Global LO: G2

28) You recently identified a molecule that you believe is a primary messenger for a signal transduction mechanism. All you know about this ligand is that it is hydrophilic. Where would you expect to find initial interaction of the molecule with its receptor?

- A) on the outer surface of the plasma membrane
- B) within the cytoplasm of the cell
- C) within the nucleus of the cell
- D) on the cytoplasmic side of the plasma membrane
- E) in endosome vesicles

Answer: A

Chapter Section: 23.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

29) Each of the following is a plant hormone *except*

- A) brassinosteroids
- B) abscisic acid.
- C) ethylene.
- D) All the above are plant hormones.
- E) None of the above are plant hormones.

Answer: D

Chapter Section: 23.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

30) Which of the following is a synthetic protein produced from engineered DNA that is used to determine whether cytosolic calcium fluxes are associated with signal transduction in a particular cell type?

- A) sildenafil
- B) ionomycin
- C) A23187
- D) fura-2
- E) a cameleons

Answer: E

Chapter Section: Key Technique

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.3

Global LO: G1

31) Which of the following would you expect to happen if you added EGF to a cell extract from which only the plasma membrane fraction has been removed?

- A) cAMP will be produced.
- B) Gs α will be activated.
- C) Adenylyl cyclase will be activated.
- D) Calcium ions will be released.
- E) None of the above will happen.

Answer: E

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

32) IP3 receptors are associated with

- A) plasma membranes.
- B) Golgi apparatus membranes.
- C) mitochondrial membranes.
- D) ER membranes.
- E) endosome membranes.

Answer: D

Chapter Section: 23.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.3

Global LO: G2

33) Dilation of blood vessels is regulated by the second messenger

- A) cAMP.
- B) Ca²⁺.
- C) IP3.
- D) CO₂.
- E) NO.

Answer: E

Chapter Section: Human Connections

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.2

Global LO: G2

34) Kinase enzymes

- A) add phosphate groups.
- B) always inhibit target molecule activity.
- C) bind tightly by wrapping around a target molecule to regulate its activity.
- D) remove phosphate groups.
- E) metabolize glucose.

Answer: A

Chapter Section: 23.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

35) Phosphatase enzymes

- A) add phosphate groups.
- B) always inhibit target molecule activity.
- C) bind tightly by wrapping around a target molecule to regulate its activity.
- D) remove phosphate groups.
- E) metabolize glucose.

Answer: D

Chapter Section: 23.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

36) Protein kinase C (PKC) activity is involved in each of the following *except*

- A) ion channel regulation.
- B) cytoskeleton changes.
- C) increase in cellular pH.
- D) secretion of proteins and other substances.
- E) PKC is involved in all of the above.

Answer: E

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

37) Each of the following hormones is an amino acid derivative *except*

- A) histamine.
- B) norepinephrine.
- C) prostaglandins.
- D) epinephrine.
- E) thyroxine.

Answer: C

Chapter Section: 23.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.2

Global LO: G2

38) Each of the following is a second messenger *except*

- A) cyclic adenosine monophosphate.
- B) inositol trisphosphate.
- C) diacylglycerol.
- D) epinephrine.
- E) calcium ions.

Answer: D

Chapter Section: 23.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

39) Each of the following is true about Ras *except*

- A) Ras is monomeric.
- B) Ras activity is inhibited by GTPase activating proteins (GAPs).
- C) Ras binds steroid hormones.
- D) Ras is activated by Sos, a guanine nucleotide exchange factor.
- E) Ras activates a cascade of phosphorylation events within the cell.

Answer: C

Chapter Section: 23.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

40) You are investigating a cell activity that you suspect is activated by a rise in intracellular calcium ions. Which of the following agents could you use to cause an increase in intracellular calcium in the cells you are studying?

- A) cameleon
- B) A23187
- C) phorbol ester
- D) sildenafil
- E) NO

Answer: B

Chapter Section: Key Technique

Bloom's Taxonomy: Application

Learning Outcome: 23.3

Global LO: G1

41) Each of the following is a characteristic of calmodulin *except*

- A) It binds two calcium ions at each end of the protein.
- B) When activated by calcium ion binding, it wraps around proteins that have calmodulin binding sites.
- C) It can activate specific kinases and phosphatases.
- D) It has a very high affinity for calcium ions, and binds calcium ions irreversibly.
- E) All are characteristics of calmodulin.

Answer: D

Chapter Section: 23.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.3

Global LO: G2

42) Each of the following is a characteristic of hormones *except*

- A) synthesis by endocrine tissue cells.
- B) secretion directly into the bloodstream.
- C) having a hormone-specific lifespan in the blood ranging from seconds to hours.
- D) interacting with specific receptors on cells in target tissues throughout the body.
- E) All are characteristics of endocrine hormones.

Answer: E

Chapter Section: 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

43) Which of the following proteins exerts direct control over the glycogen synthesis and breakdown pathways when epinephrine stimulates glycogen conversion to glucose in liver cells?

- A) glycogen phosphorylase
- B) phosphorylase kinase
- C) protein kinase A
- D) cAMP phosphodiesterase
- E) glycogen synthase

Answer: C

Chapter Section: 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

44) Insulin-stimulated increase in glucose uptake and storage in cells is promoted by activation of each of the following *except*

- A) Ras.
- B) PI 3-kinase.
- C) PTEN.
- D) Akt kinase.
- E) Glut4.

Answer: C

Chapter Section: 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

23.2 Matching Questions

Choose the item in column 2 that best matches each item in column 1.

- A) GTP
- B) potassium ions
- C) thrombin
- D) stimulates G_i activity
- E) inhibits G_i activity
- F) catalyzes formation of cAMP from ATP
- G) endocrine hormones
- H) calcium ions
- I) calmodulin
- J) PIP₂
- K) Ras
- L) GDP
- M) catalyzes formation of second messenger diacylglycerol (DAG)
- N) paracrine hormones

1) phospholipase C

Chapter Section: 23.1, 23.2, 23.3, 23.4, 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1–23.3

Global LO: G2, G7

2) precursor of diacylglycerol

Chapter Section: 23.1, 23.2, 23.3, 23.4, 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1–23.3

Global LO: G2, G7

3) second messenger associated with prevention of fertilization of an animal egg by multiple sperm cells

Chapter Section: 23.1, 23.2, 23.3, 23.4, 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1–23.3

Global LO: G2, G7

4) primary signal that activates platelets

Chapter Section: 23.1, 23.2, 23.3, 23.4, 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1–23.3

Global LO: G2, G7

5) calcium ion-activated protein that regulates other protein activity

Chapter Section: 23.1, 23.2, 23.3, 23.4, 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1–23.3

Global LO: G2, G7

6) hormones acting over long distances

Chapter Section: 23.1, 23.2, 23.3, 23.4, 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1–23.3

Global LO: G2, G7

7) monomeric G-protein

Chapter Section: 23.1, 23.2, 23.3, 23.4, 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1–23.3

Global LO: G2, G7

8) adenylate cyclase

Chapter Section: 23.1, 23.2, 23.3, 23.4, 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1–23.3

Global LO: G2, G7

9) pertussis toxin

Chapter Section: 23.1, 23.2, 23.3, 23.4, 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1–23.3

Global LO: G2, G7

10) necessary for the activation of heterotrimeric G-proteins

Chapter Section: 23.1, 23.2, 23.3, 23.4, 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1–23.3

Global LO: G2, G7

Answers: 1) M 2) J 3) H 4) C 5) I 6) G 7) K 8) F 9) E 10) A

Match the second messenger on the left with the properties on the right.

- A) opens calcium ion channels
- B) activates protein kinase C
- C) binds to calmodulin
- D) glucose mobilization
- E) smooth muscle relaxation

11) cAMP

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.2

Global LO: G2

12) IP3

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.2

Global LO: G2

13) DAG

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.2

Global LO: G2

14) calcium

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.2

Global LO: G2

15) cGMP

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.2

Global LO: G2

Answers: 11) D 12) A 13) B 14) C 15) E

23.3 Short Answer Questions

1) The α -subunit of a heterotrimeric G-protein is active when bound to _____.

Answer: GTP

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

2) EGF (epidermal growth factor) activates cell division by binding to _____.

Answer: receptor tyrosine kinases

Chapter Section: 23.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

3) Following fertilization of animal eggs, _____ are the second messengers that trigger egg activation.

Answer: calcium ions

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

4) _____ produces the second messengers IP3 and diacylglycerol (DAG).

Answer: Phospholipase C

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

5) In heterotrimeric G-proteins, the _____ subunit stimulates adenylyl kinase.

Answer: $G_{s\alpha}$

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

6) _____ are genetically engineered proteins that respond directly to cytosolic calcium fluxes by changing their fluorescence intensity.

Answer: Cameleons

Chapter Section: Key Technique

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.3

Global LO: G2

7) Types of regulators of G-protein signaling (RGS) proteins known as _____ accelerate termination of signaling by stimulating a G-protein to _____ GTP.

Answer: GAPs; hydrolyze

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

8) _____ is the messenger that regulates blood pressure by causing arterial smooth muscle relaxation.

Answer: Nitric oxide

Chapter Section: Human Connections

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.2

Global LO: G2

9) Gs α activates _____.

Answer: adenylyl cyclase

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

10) _____ nucleotides are common second messengers in animal cells.

Answer: Cyclic

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

11) The coordinated response of cells to an array of different signals from the environment is known as _____.

Answer: signal integration

Chapter Section: 23.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.2

Global LO: G2

12) Cells can regulate precisely where signaling occurs within the cell (e.g., in the yeast mating pathway) by association of the signal pathway proteins with a _____.

Answer: scaffold protein or scaffold complex

Chapter Section: 23.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

13) In smooth muscle cells, acetylcholine stimulates activation of the enzyme _____, which cleaves _____ to form the second messengers _____ and _____.

Answer: phospholipase C; PIP₂; IP₃; DAG

Chapter Section: 23.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

14) Gs α nucleotide exchange is stimulated by a _____, whereas Ras nucleotide exchange is stimulated by _____.

Answer: GPCR/EGF receptor; Sos

Chapter Section: 23.2, 23.3

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.2

Global LO: G2

15) A mutation in RTK receptor that allows the receptor to bind the ligand and pair with a wild-type receptor but blocks signal transduction is classified as a _____ mutation, whereas a mutation that activates RTK pairing signal transduction even in the absence of ligand is classified as a _____ mutation.

Answer: dominant negative; constitutively active

Chapter Section: 23.3

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1

Global LO: G2

16) PTEN is a phosphatase that removes a phosphate group from _____, thereby preventing its activation of Akt kinase.

Answer: PIP₃

Chapter Section: 23.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 23.1

Global LO: G2

23.4 Inquiry

1) What compounds or probes could you use to investigate (a) G-protein-mediated signal transduction, (b) nitric oxide signal transduction, and (c) calcium ion-dependent fluxes and effects.

Answer:

a) Nonhydrolyzable GTP analogues prolong G-protein-mediated processes because the G-protein cannot turn itself off.

b) Nitroglycerin releases NO, which can stimulate NO-dependent processes in cells.

c) Calcium ion fluxes can be measured and visualized using sensors such as cameleon that fluoresce when binding calcium ions. Calcium ion levels can be increased in cells using a calcium ionophore, which transports calcium ions across membranes.

Chapter Section: 23.2, 23.3, 23.4, 23.5

Bloom's Taxonomy: Comprehension

Learning Outcome: 23.1–23.3

Global LO: G2

2) A strain of *Drosophila* known as *dunce* produces one-half the amount of cAMP phosphodiesterase found in wild-type flies, which causes a learning defect. Another strain of *Drosophila* known as *rutabaga* produces a defective adenylyl cyclase protein (one of the AC types expressed in most cells) and also causes a learning defect. A *dunce*–*rutabaga* double mutant fly has relatively normal learning. Explain these curious observations.

Answer: cAMP accumulates to higher levels and persists longer in *dunce* fly cells because of the lower level of cAMP phosphodiesterase, which reduces cAMP levels by converting it to AMP. Elevated cAMP levels cause the *dunce* fly learning defect. *Rutabaga* fly cells make less cAMP because of the adenylyl cyclase defect. cAMP deficiency causes a learning defect. When these mutations are combined, the limited cAMP produced as a result of *rutabaga* accumulates to normal levels, because it is converted only very slowly to AMP by the low cAMP phosphodiesterase levels resulting from *dunce*.

Chapter Section: 23.2

Bloom's Taxonomy: Synthesis

Learning Outcome: 23.1

Global LO: G7, G8

3) Theophylline, a methylxanthine like caffeine, is used to treat asthma. Explain the mechanism by which a methylxanthine can alleviate asthma symptoms resulting from bronchial constriction.

Answer: Methylxanthines are cAMP phosphodiesterase (cAMP-PD) inhibitors. cAMP phosphodiesterase converts the cAMP second messenger to the inactive AMP form. By inhibiting this activity, methylxanthines raise cAMP concentrations by prolonging the existence of cAMP. cAMP stimulates PKA to phosphorylate proteins that promote bronchial smooth muscle relaxation leading to bronchodilation, thereby counteracting asthma-induced bronchoconstriction.

Chapter Section: 23.2

Bloom's Taxonomy: Application

Learning Outcome: 23.1

Global LO: G7, G8

4) Compare and contrast receptor tyrosine kinase (RTK) signaling pathways such as the EGF–Ras–MAPK pathway with serine/threonine kinase (S/TK) signaling pathways such as the

TGF β pathway with respect to (a) receptors, (b) phosphorylation targets, (c) signal transduction from the plasma membrane through the cytoplasm to the nucleus, and (d) effects on gene regulation.

Answer:

a) A single RTK contains a cytoplasmic kinase domain that autophosphorylates when primary signal bonding dimerizes the receptor. S/TK receptors that bind the primary signal associate with separate kinases that phosphorylate the ligand-binding receptor, which then binds to an activating protein.

b) In both cases, the receptors get phosphorylated. Downstream targets include the Raf–MEK–MAPK cascade kinases and transcription factors for the EGF–Ras–MAPK pathway and R-Smad for the TGF β pathway.

c) RTK recruitment of GRB2 and the Ras–GEF Sos to the membrane activates Ras, which then activates Raf to initiate the Raf–MEK–MAPK cascade. This cascade amplifies the cytosolic signal and phosphorylates MAPK, which moves into the nucleus and phosphorylates transcription factors. In the TGF β pathway, phosphorylation of R-Smad promotes complex formation with Smad4. This complex moves into the nucleus.

d) MAPK phosphorylates transcription factors, activating them to turn on specific genes. In the TGF β pathway, the Smad complex associates with other cofactors and DNA-binding proteins to regulate gene expression.

Chapter Section: 23.3

Bloom's Taxonomy: Synthesis

Learning Outcome: 23.1

Global LO: G7, G8

5) Initial attempts to culture cells demonstrated that blood serum but not blood plasma provided factors that increased the viability and proliferation of the cultured cells. Explain.

Answer: The difference between blood plasma and serum is that serum is what remains when a clot forms and is removed from the plasma. During the clotting process, platelets are activated to secrete growth factors such as PDGF, which activates fibroblast RTK receptors to stimulate cell growth and division.

Chapter Section: 23.3

Bloom's Taxonomy: Application

Learning Outcome: 23.1

Global LO: G7, G8

6) Compare and contrast GPCR and RTK receptors with respect to (a) structure—especially the transmembrane region, (b) activation mechanism, (c) initial signal transduction across the membrane.

Answer:

a) GPCRs have seven transmembrane helices with only connecting loops and the N- and C-terminal regions projecting from the membrane. RTKs have only a single transmembrane domain with large globular regions projecting outside and inside the membrane.

b) Ligand binding to a GPCR causes a conformation change in the transmembrane helices, which changes the conformation of the cytoplasmic regions to form a binding site for a G-protein.

Ligand binding causes a conformational change in the RTK extracellular domain, but with only a single transmembrane region that conformational change cannot be transmitted to the cytoplasmic domain. Instead, the extracellular domain conformation change cause two RTKs to dimerize, which brings the cytoplasmic domains close enough to autophosphorylate tyrosines to provide binding sites to recruit SH2-domain proteins.

c) Binding of G-proteins to activated GPCRs and SH2-domain proteins to activated RTKs transmits the signal into the cell.

Chapter Section: 23.2, 23.3

Bloom's Taxonomy: Analysis

Learning Outcome: 23.2

Global LO: G7, G8

Becker's The World of the Cell, 9e (Hardin/Bertoni/Kleinsmith)
Chapter 26 Cancer Cells

26.1 Multiple-Choice Questions

1) The individual credited with coining the term *cancer* is

- A) Paget.
- B) Hippocrates.
- C) Folkman.
- D) Rous.
- E) Aristotle.

Answer: B

Chapter Section: 26.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

2) Each of the following can contribute to proliferation of cancer cells *except*

- A) anchorage-independent growth.
- B) inhibition of apoptosis.
- C) density-dependent inhibition of growth.
- D) tumor-suppressor inactivation.
- E) All can contribute to immortalization.

Answer: C

Chapter Section: 26.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

3) The most frequent category of cancer is

- A) carcinoma.
- B) sarcoma.
- C) lymphoma.
- D) leukemia.
- E) None of the above categories is the most frequent.

Answer: A

Chapter Section: 26.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

4) Each of the following is a way cancer cells avoid growth and invasion restriction imposed by the immune system *except*

- A) They produce a heterogeneous group of cells.
- B) They respond to density-dependent growth factors.
- C) They produce molecules that inhibit T lymphocyte function.
- D) They surround themselves with dense supporting tissue.
- E) They express low levels of surface antigens.

Answer: B

Chapter Section: 26.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

5) Compared to a normal cell, a cancer cell generally divides

- A) ten times faster than a normal cell.
- B) four times faster than a normal cell.
- C) two times faster than a normal cell.
- D) at the same rate as a normal cell.
- E) much slower than a normal cell.

Answer: D

Chapter Section: 26.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

6) Judah Folkman performed an experiment in which he found that small tumors floating in the anterior chamber of the eye grew little, whereas a similar tumor implanted directly onto the iris grew to thousands of times its original size. These results supported his idea that continued tumor growth is dependent on requirement of tumor growth for

- A) continued exposure to a carcinogen.
- B) evasion of apoptosis.
- C) angiogenesis.
- D) metastasis.
- E) a stepwise series of gene mutations.

Answer: C

Chapter Section: 26.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

7) A cell that has acquired the capacity to grow suspended in solution is

- A) infected with a virus.
- B) immortalized.
- C) apoptotic.
- D) anchorage-independent.
- E) density-dependent.

Answer: D

Chapter Section: 26.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

8) Polycyclic aromatic hydrocarbons in tobacco smoke are potent carcinogens because they preferentially bind to, and trigger base substitutions, in the

- A) *Ras* gene.
- B) *p53* gene.
- C) *Raf* gene.
- D) *PTEN* gene.
- E) *EGF* receptor gene.

Answer: B

Chapter Section: 26.3

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

9) The type of protein that binds cells to the extracellular matrix is a(n)

- A) integrin.
- B) thrombospondin.
- C) growth factor.
- D) matrix metalloproteinase.
- E) Ras protein.

Answer: A

Chapter Section: 26.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

10) The expression of which of the following contributes to development of cancer?

- A) tumor suppressor genes
- B) oncogenes
- C) proto-oncogenes
- D) apoptosis genes
- E) the *p53* gene

Answer: B

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

11) Cancer cell aneuploidy (the gain or loss of chromosomes) is most likely caused by a defect in

- A) *RB*.
- B) *p53*.
- C) *Myc*.
- D) *BCL2*.
- E) *Mad*.

Answer: E

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

12) Organ transplant patients often develop cancers at rates higher than normal. This observation supports the idea that cancer formation is dependent on

- A) anchorage-independence of growth.
- B) evasion of immune surveillance.
- C) multistep mutations.
- D) angiogenesis.
- E) increased motility of cancer cells.

Answer: B

Chapter Section: 26.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

13) A new compound developed in the lab was tested as a mutagen. When the Ames test was performed, the results were numerous bacterial colonies following exposure. This result means the chemical

- A) is safe.
- B) is radioactive.
- C) would be a great candidate to use for chemotherapy.
- D) is a mutagen and also may be a carcinogen.
- E) causes apoptosis.

Answer: D

Chapter Section: 26.3

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

14) In the Ames test, gene mutation restores the ability of a bacterium to synthesize

- A) epidermal growth factor.
- B) histidine.
- C) a polycyclic aromatic hydrocarbon.
- D) a T lymphocyte inhibitor.
- E) a DNA repair enzyme.

Answer: B

Chapter Section: 26.3

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

15) Telomere maintenance in cancer cells is the result of

- A) natural cell aging.
- B) active telomerase activity.
- C) the exchange of DNA between chromosomes.
- D) viral infection.
- E) Both B and C result in telomere maintenance.

Answer: E

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

16) Carcinogens can damage DNA by each of the following mechanisms *except*

- A) by binding to DNA and directly disrupting base pairing.
- B) by generating cross-links between DNA strands.
- C) by hydroxylating DNA bases.
- D) by causing breaks in one or both strands.
- E) All the above are mechanisms by which carcinogens can damage DNA.

Answer: E

Chapter Section: 26.3

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

17) Each of the following is true with regard to cancer cells *except*

- A) growth does not require growth factors.
- B) the cell cycle may be disrupted.
- C) most cancer cells age and die after 50 divisions.
- D) growth may proceed despite high levels of DNA damage.
- E) the apoptosis pathway may be blocked.

Answer: C

Chapter Section: 26.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

18) Judah Folkman used which experimental model system to explore the relationship between tumor growth and angiogenesis?

- A) a rabbit eye
- B) the colon
- C) bone marrow
- D) mouse skin
- E) in vitro cells

Answer: A

Chapter Section: 26.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

19) A tumor is actively producing both VEGF and FGF, factors that stimulate angiogenesis, but no new network of blood vessels to the tumor has been established in spite of the fact that blood vessels are nearby. This observation suggests that most likely

- A) the cancer cell versions of the VEGF and FGF are not functional.
- B) only VEGF or EGF is active but not both.
- C) p53 is overexpressed.
- D) angiostatin is present in greater concentration.
- E) the tumor has been established by metastasis of a cell from a distant tumor.

Answer: D

Chapter Section: 26.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

20) Histone modifications that play a role in silencing expression of tumor suppressor genes are influenced directly by

- A) microRNAs.
- B) epidermal growth factor signaling.
- C) p53 activity.
- D) carcinogens.
- E) APC activity.

Answer: A

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

21) UV light generates which of the following as a "signature" mutation in a gene?

- A) DNA hydroxylation
- B) a single-strand break
- C) pyrimidine dimer formation
- D) cross-links between DNA strands
- E) changes in DNA methylation

Answer: C

Chapter Section: 26.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

- 22) Each of the following is true regarding cancer cell metastasis to secondary sites *except*
- A) cancer cells may multiply in regional lymph nodes if they enter via lymphatics.
 - B) many cells from the primary mass may enter circulation.
 - C) all cancer cells circulating in the bloodstream are identical.
 - D) few cancer cells survive in the bloodstream.
 - E) the population of cancer cells is heterogeneous.

Answer: C

Chapter Section: 26.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

- 23) Precarcinogens are themselves not carcinogens but may be activated to become a carcinogen in the liver by the activity of

- A) PTEN.
- B) Ras.
- C) EGF.
- D) Cytochrome P450.
- E) APC.

Answer: D

Chapter Section: 26.3

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

- 24) The promotion phase of carcinogenesis is enhanced by

- A) prolonged or repeated exposure to foreign or natural agents that stimulate cell proliferation.
- B) inactivation of Ras activity.
- C) natural selection of cells exhibiting enhanced growth rate and invasive properties.
- D) a single exposure to a carcinogen.
- E) a single base-pair change in genomic DNA.

Answer: A

Chapter Section: 26.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

25) The branch of medicine that investigates the frequency and distribution of human diseases is

- A) immunology.
- B) physiology.
- C) pathology.
- D) epidemiology.
- E) gene therapy.

Answer: D

Chapter Section: 26.3

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

26) The first demonstration that cancer could be caused by a virus involved

- A) Rous sarcoma virus in chickens.
- B) Epstein–Barr virus in Burkitt's lymphoma.
- C) hepatitis B in liver cancer.
- D) HIV in human Kaposi's sarcoma.
- E) human papilloma virus in cervical cancer.

Answer: A

Chapter Section: 26.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

27) Chronic inflammation caused by infectious agents can increase the chance of cancer because immune cells fighting the infection

- A) induce conversion of the *RAS* proto-oncogene to the oncogene in neighboring cells.
- B) secrete metalloproteinases that digest holes in the surrounding extracellular matrix.
- C) secrete phorbol esters.
- D) secrete an oncogenic form of a growth factor.
- E) produce mutagenic oxygen-free radicals.

Answer: E

Chapter Section: 26.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

28) The viral oncogene associated with the Rous sarcoma virus is the

- A) *RAS* gene.
- B) *p53* gene.
- C) *ERBB2* gene.
- D) *v-sis* gene.
- E) *src* gene.

Answer: E

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

29) Each of the following is a mechanism by which proto-oncogenes are converted to oncogenes *except*

- A) point mutation.
- B) gene amplification.
- C) translocation.
- D) deletion.
- E) histone methylation.

Answer: E

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

30) Oncogenes encode proteins in each of the following categories *except*

- A) growth-factor receptors.
- B) nonreceptor protein kinases.
- C) transcription factors.
- D) cytochrome P450 enzymes.
- E) growth factors.

Answer: D

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

31) Fusion of a cancer cell and a normal cell using polyethylene glycol yields a hybrid cell that behaves like a normal cell because

- A) of recombination between cancer cell and normal cell chromosomes.
- B) of dilution of carcinogens in the hybrid cell cytoplasm.
- C) of the presence of normal cell tumor suppressor genes and gene products in the hybrid cell.
- D) exposure to polyethylene glycol inhibits expression of oncogenes.
- E) the normal cell DNA repair machinery reverses all cancer cell gene mutations.

Answer: C

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

32) Each of the following is a category into which proteins encoded by oncogenes fit *except*

- A) growth factors.
- B) signal receptors.
- C) plasma membrane-associated GTP-binding proteins.
- D) cell cycle regulators.
- E) P450 cytochromes.

Answer: E

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

33) In some forms of breast cancer, estrogen is a(n)

- A) initiator.
- B) promoter.
- C) suppressor.
- D) inhibitor.
- E) mutagen.

Answer: B

Chapter Section: 26.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

34) Children with hereditary retinoblastoma inherit a chromosome 13 from one parent that has which of the following defects in the *RB* gene?

- A) insertion
- B) duplication
- C) inversion
- D) point mutation
- E) deletion

Answer: E

Chapter Section: 26.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

35) Individuals can inherit a defective *APC* gene on chromosome 5 from one parent that makes them susceptible to developing familial adenomatous polyposis. In which cells is this deletion present in affected individuals?

- A) somatic cells
- B) germ cell precursors
- C) colon polyp cells
- D) colon epithelial cells
- E) all cells

Answer: E

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

36) Most hereditary forms of breast and ovarian cancer arise in women who inherit a mutant copy of either the *BRCA1* or *BRCA2* gene, each of which encodes a proteins involved directly in

- A) inhibiting apoptosis.
- B) repairing DNA double-strand breaks.
- C) passing through the G1 restriction point.
- D) a spindle assembly checkpoint.
- E) activating Rho proteins to increase cell motility.

Answer: B

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

37) The normal *RB* gene product functions as

- A) a transcription factor.
- B) a signaling pathway protein.
- C) an apoptosis pathway protein.
- D) a cell-surface receptor.
- E) a cell-cycle protein.

Answer: E

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

38) Each of the following is a main cause of cancer *except*

- A) carcinogens.
- B) radiation.
- C) infectious agents.
- D) heredity.
- E) All are causes of cancer.

Answer: E

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

39) Epigenetic changes in gene expression that can lead to cancer by silencing tumor suppressor genes are mediated by

- A) hereditary mutations in gene base sequence.
- B) spontaneous mutations in gene base sequence.
- C) DNA methylation.
- D) DNA double-strand breaks.
- E) pyrimidine dimer formation.

Answer: C

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

40) Which of the following viruses has *not* been associated with a particular type of cancer?

- A) influenza virus
- B) hepatitis B virus
- C) Epstein–Barr virus
- D) human papilloma virus
- E) hepatitis C virus

Answer: A

Chapter Section: 26.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

41) A physician using a microscope to examine suspect cells in biopsy tissue would expect to find each of the following characteristics in metastatic cancer cells *except*

- A) poorly differentiated cells.
- B) many dividing cells.
- C) perfectly spherical nuclei.
- D) a poorly defined tumor boundary.
- E) a high nucleus to cytoplasmic volume ratio.

Answer: C

Chapter Section: 26.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.2

Global LO: G2

42) Each of the following is a cancer a cancer screening procedure *except*

- A) Pap smear.
- B) mammography.
- C) PSA test.
- D) colonoscopy.
- E) All are screening procedures.

Answer: E

Chapter Section: 26.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.2

Global LO: G2

43) Different chemotherapeutic drugs can do each of the following *except*

- A) inhibit metabolic pathways required for DNA synthesis.
- B) chemically cross-link DNA strands.
- C) inhibit DNA topoisomerases required for DNA replication.
- D) disrupt spindle microtubules.
- E) Chemotherapeutic drugs can do all of the above.

Answer: E

Chapter Section: 26.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.2

Global LO: G2

44) Immunotherapy approaches for treating cancer include each of the following *except*

- A) inserting bacteria to provoke a local immune response after removing a tumor.
- B) injecting humanized antibodies that recognize cancer cell surface proteins.
- C) stimulating immune system response with interleukin-2 or interferon alpha.
- D) introducing cancer cell antigens to stimulate immune system response to selectively attack cancer cells.
- E) All of the above are immunotherapy approaches for treating cancer.

Answer: E

Chapter Section: 26.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.2

Global LO: G2

45) The anti-angiogenic drug Avastin works by binding to and inhibiting the activity of

- A) EGF.
- B) VEGF.
- C) estrogen.
- D) IL-2.
- E) Ras.

Answer: B

Chapter Section: 26.5

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.2

Global LO: G2

26.2 Matching Questions

Match the infectious agent on the left with the appropriate cancer type on the right.

- A) stomach cancer
- B) Burkitt's lymphoma
- C) liver cancer
- D) cervical cancer
- E) bladder cancer

1) Epstein–Barr virus
Chapter Section: 26.3
Bloom's Taxonomy: Knowledge
Learning Outcome: 26.1
Global LO: G2

2) hepatitis B virus
Chapter Section: 26.3
Bloom's Taxonomy: Knowledge
Learning Outcome: 26.1
Global LO: G2

3) human papilloma virus
Chapter Section: 26.3
Bloom's Taxonomy: Knowledge
Learning Outcome: 26.1
Global LO: G2

4) *Helicobacter pylori*
Chapter Section: 26.3
Bloom's Taxonomy: Knowledge
Learning Outcome: 26.1
Global LO: G2

5) parasitic flatworms
Chapter Section: 26.3
Bloom's Taxonomy: Knowledge
Learning Outcome: 26.1
Global LO: G2

Answers: 1) B 2) C 3) D 4) A 5) E

Match the gene on the left with the gene category on the right. Note choices on the right may be used more than once or not at all.

- A) proto-oncogene
- B) low expression in invasive cells
- C) tumor suppressor gene
- D) oncogene

6) *BRCA1*

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

7) *E-cadherin*

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

8) *p53*

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

9) *ERBB2*

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

10) *Cdk4*

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

11) *c-myc*

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

12) *KRAS*

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

13) *RB*

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

14) *v-src*

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

Answers: 6) C 7) B 8) C 9) A 10) D 11) D 12) D 13) C 14) D

26.3 Short Answer Questions

1) _____ are cancers that originate in epithelial cells and comprise 90% of all cancers.

Answer: Carcinomas

Chapter Section: 26.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

2) A _____ is a cancer of blood cells in which the cells proliferate in the bloodstream.

Answer: leukemia

Chapter Section: 26.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

3) Some cancer cells produce the enzyme _____, which protects the ends of chromosomes during multiple rounds of cell division.

Answer: telomerase

Chapter Section: 26.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

4) The enzyme _____ breaks down the extracellular matrix to facilitate angiogenesis.

Answer: matrix metalloproteinase

Chapter Section: 26.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

5) _____ is the direct migration and penetration of cancer cells into neighboring tissues.

Answer: Invasion

Chapter Section: 26.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

6) Highly invasive cancer cells usually express less _____, a key adhesion molecule.

Answer: E-cadherin

Chapter Section: 26.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

7) Cancer cells secrete an activator of the enzyme _____ to degrade basal lamina and extracellular matrix and activate matrix metalloproteinases.

Answer: plasmin

Chapter Section: 26.2

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

8) Epidemiological studies have revealed that cancers arise with differing frequencies in different parts of the world because _____ and _____ trigger different DNA mutations associated with particular types of cancers.

Answer: environmental agents; lifestyle factors

Chapter Section: 26.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

9) An individual can inherit a gene in which expression has been altered by an _____ change with no mutation of the gene sequence.

Answer: epigenetic

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

10) Finding the _____ virus in chickens established the first association between viral infection and cancer.

Answer: Rous sarcoma

Chapter Section: 26.3

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

11) Epigenetic effects on gene expression may occur through covalent modifications of _____ and _____.

Answer: DNA; histones

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

12) Overexpression of _____, which bind to and silence mRNAs, may lead to the development of cancer.

Answer: micro-RNAs

Chapter Section: 26.4

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

13) Three or more spindle poles present during mitosis of actively dividing cancer cells causes _____.

Answer: aneuploidy

Chapter Section: 26.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

14) Precarcinogens are themselves not carcinogens but may be modified in the liver to become a carcinogen by a process known as _____.

Answer: carcinogen activation

Chapter Section: 26.3

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

15) Tumor progression by clonal expansion is based on the idea that the original population of cancer cells is subjected to _____ selection.

Answer: Darwinian

Chapter Section: 26.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

16) Nude mice are useful for investigating cancer cells because the nude mice lack a functional

_____ system.

Answer: immune

Chapter Section: 26.1

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

17) As in development of normal tissue such as skin, a tumor may contain cancer _____ cells that generate large numbers of descendants.

Answer: stem

Chapter Section: 26.1

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

18) In addition to cleaving extracellular matrix structural components, plasmin can facilitate metastasis by cleaving precursors to activate matrix _____.

Answer: metalloproteinases

Chapter Section: 26.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

19) For approximately two-thirds of human cancers, the rates of metastasis to different organs can be explained by _____ patterns.

Answer: blood-flow

Chapter Section: 26.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

20) Although not predicted by blood-flow patterns, prostate cancer cells commonly metastasize to bone, because bone cells produce specific _____.

Answer: growth factors

Chapter Section: 26.2

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

21) Tumor suppressor genes can be split into the two categories of _____ and _____ genes.

Answer: gatekeeper; caretaker

Chapter Section: 26.4

Bloom's Taxonomy: Knowledge

Learning Outcome: 26.1

Global LO: G2

26.4 Inquiry

1) Epidemiologists at the CDC in Atlanta, GA, have called you in to investigate an increased incidence of leukemia in a remote region of Africa. Shortly after you arrive, an RNA virus was isolated from patients and found to induce leukemia in mice. At this point, you don't know the viral gene product or the mechanism that causes development of leukemia. One of your lab assistants has isolated a variant of the virus that is missing 20% of the genome of the original virus and does not induce leukemia in mice.

a) How would you determine which portion of the genome is associated with the development of leukemia?

b) How would you determine the viral product or mechanism associated with the development of leukemia?

Answer:

a) Sequence the viral DNA from each virus strain and compare the sequence. Mutate the suspected gene in the normal virus and assay its effect on causing leukemia in mice.

b) Look within the viral genome for sequences of a known oncogene (such as *src*), or test infected cells for viral genome insertion near or within a cellular proto-oncogene.

Chapter Section: 26.3

Bloom's Taxonomy: Comprehension

Learning Outcome: 26.1

Global LO: G2

2) Cancer-causing mutations in genes can have different effects on the protein products expressed.

a) What type of mutation would be dominant in the development of cancer? Why?

b) What type of mutation would be expressed as a recessive trait in the development of cancer? Why?

c) Based upon your answers to parts (a) and (b), how would you treat these situations using a gene therapy approach?

Answer:

a) A mutation that converts a proto-oncogene into an oncogene would be dominant because expression of the oncogenic form of the protein would be dominant over any normal cellular regulation of the normal protein activity.

b) A mutation in a tumor suppressor gene that destroys its product's expression or activity would be recessive because expression of protein from a normal allele would supply enough protein to prevent uncontrolled cell division.

c) Various types of gene therapy could be used to delete the oncogene or provide a good copy of the tumor suppressor gene.

Chapter Section: 26.4

Bloom's Taxonomy: Synthesis

Learning Outcome: 26.1, 26.2

Global LO: G7, G8

3) The more common types of cancers evade immune system destruction by becoming less detectable and by actively overcoming the immune system: (a) give an example of each, and (b) propose an experiment that could distinguish between these possibilities.

Answer:

a) Tumor cells are heterogeneous and contain cells that lack or express low levels of surface antigens recognized by immune cells. Cancer cells can actively overcome the immune system by producing molecules that kill or inhibit the function of T lymphocytes. Tumors also can surround themselves with dense supporting tissue that shields them from immune attack, and some cancer cells divide so quickly that the immune system cannot kill them fast enough to keep tumor growth in check.

b) One way to assess antigen expression is to use labeled antibodies to known antigens to stain the tumor cells. One way to assess the possibility that the tumor cells are secreting T cell-inhibiting molecules is to assay fluid extracted from a tumor or culture of the tumor cells for effects on T lymphocyte activity.

Chapter Section: 26.2

Bloom's Taxonomy: Synthesis

Learning Outcome: 26.2

Global LO: G7, G8

4) You have isolated and sequenced a region of DNA that appears to have caused tumor formation. For what types of mutations would you look to understand how that DNA causes the cancer and what is the characteristic of each type of mutation you might find?

Answer:

(a) Point mutation: A single base-pair change substitution could change an amino acid in an encoded protein that activates the activity of an oncogene product or inactivates a tumor suppressor gene product.

(b) Local DNA rearrangement: The DNA may have an insertion, deletion, inversion, or transposition that converts a proto-oncogene to an oncogene or disrupts a tumor suppressor gene.

(c) Gene amplification: creates multiple copies of a gene, which causes overactivity of the gene product, because it is overexpressed.

(d) Chromosomal translocation: The DNA could have moved to a different part of the same chromosome or to a different chromosome. This may fuse two genes together to form an oncogene encoding an abnormal protein, or it may place a proto-oncogene next to a highly active gene, thereby causing the translocated proto-oncogene to become more active.

(e) Insertional mutagenesis: Like translocation, movement of the DNA may integrate it near a proto-oncogene, stimulating its expression, or into a tumor suppressor gene, preventing expression of a functional protein.

Chapter Section: 26.4

Bloom's Taxonomy: Analysis

Learning Outcome: 26.1

Global LO: G7

5) You have isolated two novel compounds, Compound X and Compound Y, which you suspect might be carcinogenic in some way. Based on their chemical characteristics, you hypothesize that Compound X is a cancer initiator and Compound Y is a cancer promoter. Design an experiment to test your hypothesis.

Answer: Treat a laboratory animal with one dose of X followed at a later time by treatment with Y. This protocol should result in cancer if X is an initiator and Y is a promoter. Control experiments should include treating animals with only X and only Y, and treating animals with Y first followed by X. None of these protocols should be very effective at causing cancer.

Chapter Section: 26.1

Bloom's Taxonomy: Synthesis

Learning Outcome: 26.1

Global LO: G7, G8